Selective Prediction and the Persistence Illusion: A Diagnostic Decomposition of VIX Regime Classification

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Abstract

We present a diagnostic protocol for evaluating selective prediction frameworks on autocorrelated classification labels, comprising six components: coverage-matched baseline comparison, persistence quantification, regime-transition accuracy audit, SHAP feature attribution, sequence-model (LSTM) comparison, and cross-task replication. Applied to VIX regime classification over 5,000 trading days (2006–2026) and replicated on S&P 500 trend-regime classification, the protocol shows that confidence-based selective prediction manufactures apparent skill by concentrating predictions on regime-stable days. Key findings: (1) the confidence filter achieves 0% accuracy on every calm→high volatility transition it covers, and explicit cost-weighting up to 20× does not rescue this—the blindness is informational, not objective-function-driven; (2) at $N \geq 10$, Random Forest selective prediction and coverage-matched persistence make *identical* predictions on every jointly covered day (McNemar $b = c = 0$), so the accuracy gap is attributable entirely to coverage composition, not multi-feature skill; (3) a HAR-style logistic regression on three VIX features matches or exceeds the 36-feature Random Forest in point estimate at four of five horizons, with no gap reaching statistical significance under moving-block bootstrap; (4) at $N = 25$, balanced accuracy collapses to 50%, confirming that headline selective accuracy is an artifact of class imbalance; and (5) under any moderate false-negative cost asymmetry ($C \geq 5$), persistence at matched coverage dominates RF selective prediction on cost-weighted utility at $N = 10$. Three supporting analyses confirm the persistence-illusion mechanism: SHAP attribution shows VIX technical features account for 71.9% of model prediction mass; a direct decomposition shows 99–100% of correct covered predictions are same-regime calls at every horizon; and a two-layer LSTM with 20-day temporal memory also fails on calm→high transitions (35% covered accuracy, below random chance), showing that architectural richness does not resolve the information constraint. The full pattern replicates on S&P 500 trend regimes and yield curve inversion (persistence ratio 75.5×; RF−Persist = −53 pp), establishing the persistence illusion as a general property of selective prediction on serially correlated labels whose severity scales with the persistence ratio. A synthetic positive control confirms that the protocol has power: injecting a genuine oracle alarm feature produces RF−Persist = +8.7 pp with calm→high accuracy of 54.9%, satisfying all diagnostic criteria.

Contents

1 Introduction

Volatility is one of the most important dimensions of financial market risk. The CBOE Volatility Index (VIX), often described as the market's "fear gauge," summarizes option-implied expectations of S&P 500 volatility over the next 30 days and is widely used by investors, risk managers, and policymakers to monitor uncertainty in equity markets [Whaley, 2009]. When VIX rises above 20—a level widely adopted as a practical divide between normal and elevated-fear environments [Whaley, 2009]—markets are generally considered to be in a high-volatility regime characterized by elevated fear, widening credit spreads, and increased probability of large equity drawdowns. Episodes such as the February 2018 "Volmageddon" spike and the COVID-19 crash of March 2020 illustrate the costs of failing to anticipate these regime transitions: institutions caught under-hedged can face rapid, severe losses. A reliable model for predicting high-volatility regimes would therefore have direct practical value for risk management, hedging, and portfolio construction.

Despite its importance, VIX regime prediction is a difficult problem. Volatility is noisy, mean-reverting, and driven by a complex mixture of market sentiment, equity conditions, interest rates, and sudden exogenous shocks. Classical approaches such as GARCH-family models [Bollerslev, 1986] and Markov-switching frameworks [Hamilton, 1989] have provided important foundations, but often assume specific parametric structures that may not hold in practice. Machine learning methods offer flexibility in modeling nonlinear interactions across large feature sets, but introduce their own challenges: temporal leakage, calibration, and—critically for regime classification—the difficulty of distinguishing genuine predictive skill from the strong autocorrelation inherent in VIX.

A further challenge is that not all trading days provide equally useful predictive information. The selective prediction literature—which studies classifiers that may abstain from uncertain predictions [Chow, 1970, Bartlett and Wegkamp, 2008, Geifman and El-Yaniv, 2017]—offers a principled solution: reserve predictions for high-confidence cases, trading coverage for reliability on predicted days. On autocorrelated label sequences such as VIX regimes, however, selective prediction can manufacture the appearance of skill by preferentially covering easy regime-stable days.

This paper develops a confidence-based selective prediction framework for VIX regime classification and uses it as a lens to examine where accuracy gains come from and what they are worth. Two ensemble classifiers—Random Forest (RF) and Histogram Gradient Boosting (HGB)—are trained on 36 features and abstain when calibrated probability is close to 0.5. The headline accuracy figures (90–93% for RF at $\tau = 0.25$) are striking, but a systematic decomposition reveals that they constitute a diagnostic, not an achievement.

The primary scientific contribution of this paper is the diagnostic decomposition protocol itself. This protocol operationalizes a methodological critique implicit in the machine learning evaluation literature—that confidence-based selective prediction on autocorrelated labels manufactures apparent accuracy by concentrating predictions on regime-stable days—and makes it formally testable through six reproducible components. Each component isolates a distinct failure mode: (i) coverage-matched baseline comparison tests whether simpler models achieve identical accuracy at identical coverage; (ii) regime-transition audit establishes whether accuracy gains concentrate on stable versus transitional days; (iii) persistence quantification measures the fraction of correct predictions attributable to same-regime calls; (iv) SHAP attribution confirms that feature importance is dominated by autocorrelation signals; (v) sequence-model comparison (LSTM) tests whether architectural complexity resolves the information constraint; and (vi) cross-task replication establishes generalizability across autocorrelated targets. A single accuracy table cannot distinguish genuine predictive skill from persistence exploitation; this protocol can. Its application to VIX regime classification, S&P 500 trend regimes, and yield curve inversion produces a consistent finding: at every component and across all three tasks, the apparent accuracy advantage of selective prediction reduces to autocorrelation. A synthetic positive control confirms the protocol has power to identify genuine skill when it exists.

The main findings are as follows. First, a HAR-style logistic regression using only three features (current VIX level, 5-day VIX mean, 20-day VIX mean) matches or exceeds the full 36-feature RF in point estimate at four of five horizons at matched coverage; the HAR-minus-RF gap at $N = 10$ is +1.1 percentage points (pp; HAR leads), with bootstrap 95% CI $[-1.08, +4.18]$, so the advantage is a pattern, not a confirmed result, and the 33 additional features contribute negligibly. Second, a coverage-matched naive persistence rule closely tracks RF performance at all horizons; the RF-minus-persistence gap at $N = 10$ (+1.88 pp) has bootstrap 95% CI $[-1.11, +4.17]$; no horizon is statistically significant, and a McNemar test on jointly covered days finds $b = c = 0$ at $N \geq 10$: RF and persistence make identical predictions on every shared day, confirming the gap is attributable to coverage composition. Third, the confidence filter covers 87% of calm-persisting days at $N = 5$

but yields 0% accuracy on every calm→high transition it covers; explicit cost-weighting up to 20× does not overcome this, confirming the failure is informational rather than objective-function-driven. Fourth, at $N = 25$, balanced accuracy collapses to 50% for both RF and HGB, confirming that headline selective accuracy at the longest horizon is an artifact of class imbalance rather than predictive skill. Fifth, at $N = 10$, persistence at matched coverage dominates RF selective prediction on cost-weighted utility whenever false negatives are more costly than false alarms ($C \geq 5$). An exploratory post-hoc observation also merits mention: the model's abstention rate before a spike is 2.3× its overall rate at $N = 5$ (rate gap = +0.307; bootstrap 95% CI $[+0.162, +0.443]$), though VIX proximity to the threshold explains most of this signal and the finding requires out-of-sample confirmation.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the data. Section 4 presents the methodology. Section 5 reports empirical results, including naive-baseline comparison (Section 5.2), persistence quantification (Section 5.3), regime-transition case studies (Section 5.4), statistical significance (Section 5.6), LSTM comparison (Section 5.5), SHAP attribution (Section 5.7), and cost-weighted utility (Section 5.8). Section 6 presents a synthetic positive control demonstrating that the protocol correctly identifies genuine predictive skill when an oracle alarm signal is injected. Section 7 replicates the full diagnostic on S&P 500 trend regimes and yield curve inversion, establishing the persistence illusion as a general structural property that scales with the persistence ratio. Section 8 concludes. Robustness checks (threshold sensitivity, VIX threshold robustness, sustained labels, walk-forward validation, calibration quality, permutation importance) appear in the Appendix.

2 Related Work

2.1 Volatility Forecasting and Regime Modeling

The modeling of financial volatility has a long history in econometrics. Bollerslev [1986] introduced the GARCH model, which captures the tendency of volatility to cluster over time by modeling conditional variance as a function of past squared residuals. GARCH-family models have since been extended in numerous directions and remain a standard benchmark for volatility forecasting. Hamilton [1989] introduced Markov-switching models, in which a time series is assumed to switch between discrete latent states according to a Markov chain. This framework provides a natural model for regime transitions and has been widely applied to financial and macroeconomic time series. Guidolin and Timmermann [2007] extended regime-switching ideas to multivariate portfolio allocation settings, demonstrating that regime-dependent models can outperform single-state specifications in out-of-sample asset allocation.

A complementary strand of work focuses on realized volatility constructed from high-frequency intraday returns. Corsi [2009] proposed the Heterogeneous Autoregressive (HAR) model, which captures the long-memory properties of realized volatility through a simple additive structure of daily, weekly, and monthly volatility components. The HAR model has become a standard benchmark for realized volatility forecasting, and its strong performance motivates including a HAR-style logistic regression as a baseline in this paper.

In the VIX-specific literature, Whaley [2009] provided an analysis of the construction and behavior of the VIX index across market cycles, documenting its role as a fear indicator and its asymmetric behavior in market downturns. Fernandes et al. [2014] studied the predictability of VIX using a variety of econometric models and found that simple autoregressive specifications perform competitively against more complex alternatives, though nonlinear models show gains at longer horizons. This paper replicates and extends that finding: simple HAR-style models are competitive with 36-feature ensemble classifiers on the binary regime classification task.

2.2 Machine Learning for Financial Prediction

Machine learning methods have increasingly been applied to financial forecasting. Ensemble methods such as Random Forests [Breiman, 2001] and Gradient Boosting are particularly well-suited to financial prediction because they can model nonlinear interactions among large feature sets without requiring distributional assumptions. These models have been applied to equity return prediction, credit risk assessment, and volatility estimation—including VIX forecasting [Fernandes et al., 2014]—often outperforming linear alternatives when feature sets include technical indicators, cross-asset signals, and macroeconomic variables.

A key methodological challenge in applying machine learning to financial data is avoiding data leakage and respecting the temporal ordering of observations. Random cross-validation on time series data allows future observations to inform model training, producing overly optimistic estimates of out-of-sample performance. This paper addresses this by using chronological train/test splitting, `TimeSeriesSplit` cross-validation during calibration, and walk-forward validation as a robustness check on fold stability (Appendix E)—consistent with best practices in applied financial machine learning. The primary performance estimates use a single chronological 80/20 holdout.

2.3 Selective Prediction and the Reject Option

The reject option in classification allows a model to abstain from making a prediction when its confidence is insufficient. Chow [1970] first formalized the optimal reject option, showing that a classifier can achieve a favorable tradeoff between recognition error and abstention rate under a cost framework. Bartlett and Wegkamp [2008] studied classification with a reject option using hinge loss and established theoretical generalization bounds for such classifiers. Geifman and El-Yaniv [2017] demonstrated that selective prediction can substantially improve reliability in modern deep learning settings, and proposed using softmax confidence scores as a practical abstention criterion.

In financial applications, the reject option is particularly natural. A risk model that provides no signal on ambiguous days is often more useful than one that forces a low-confidence prediction, since acting on an uncertain signal may itself introduce risk. However, on autocorrelated series such as VIX regimes, abstention can be selection-driven in a way that manufactures accuracy without genuine predictive skill—the central diagnostic this paper examines.

3 Data

3.1 Market Data

Daily price data are obtained from Yahoo Finance for eight market series: the CBOE Volatility Index (VIX), the 3-month VIX term structure index (VIX3M), the CBOE 9-Day Volatility Index (VIX9D), the S&P 500 Index, gold futures (GC=F), the 10-Year Treasury Yield (TNX), the U.S. Dollar Index (DXY), and the CBOE SKEW Index. VIX9D is downloaded to keep the raw data complete but is not used as a model feature; it is excluded from the engineered feature set at the preprocessing stage. These variables capture implied volatility and its term structure, equity market conditions, safe-haven demand, interest rate movements, dollar strength, and the market's perception of tail risk, respectively. Although raw VIX data is available from January 1990, VIX3M data begins in July 2006; because VIX3M anchors the term spread feature, the effective sample starts on July 17, 2006.

3.2 Macroeconomic Data

Two macroeconomic variables are obtained from the Federal Reserve Economic Data (FRED) API: the Federal Funds Rate and the 10-Year minus 2-Year Treasury yield curve spread (T10Y2Y). The Federal Funds Rate target is set at FOMC meetings (approximately eight per year) and is forward-filled to daily frequency by holding the prevailing rate constant between meeting dates. This forward-fill introduces a mild lag: the value used on a given trading day reflects the most recent meeting outcome, which FRED may publish with a delay of up to one business day. In practice, FOMC decisions are immediately priced into markets, so this lag is unlikely to introduce lookahead bias.

3.3 Label Construction

The prediction task is binary regime classification. For a prediction horizon N, the label is defined as:

$$\text{LABEL}_{N,t} = \begin{cases} 1 & \text{if } \text{VIX}_{t+N} \geq 20 \\ 0 & \text{if } \text{VIX}_{t+N} < 20 \end{cases} \tag{1}$$

The label is forward-shifted N days to ensure that all features are observed at time t and the outcome is measured at time $t + N$, thereby eliminating lookahead bias. Horizons $N \in \{5, 10, 15, 20, 25\}$ trading days are evaluated.

A *sustained-high* label is also constructed as a stricter alternative:

$$\text{SUSTAINED\_LABEL}_{N,t} = \begin{cases} 1 & \text{if } \text{VIX}_{t+k} \geq 20 \text{ for all } k \in \{1, \ldots, N\} \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

This requires VIX to remain in the high-volatility regime on every day from $t + 1$ through $t + N$, rather than only at the endpoint. Sustained labels filter out transient single-day crossings; results appear in Appendix D.

3.4 Class Distribution and Train/Test Split

The full feature dataset covers 5,000 trading days from July 17, 2006 to May 29, 2026. Rolling and momentum features produce leading NaN values for approximately the first 20 trading days; forward-looking labels are undefined for the last N days of the sample. Observations with any missing feature or label value are dropped prior to splitting, so the effective sample size varies slightly by horizon. Data are split chronologically, with the first 80% of the cleaned sample used for training and the remaining 20% for testing, preserving strict temporal ordering.

Table 1: Class distribution (fraction of days with VIX $\geq$ 20) by dataset split.

Split	VIX $\geq$ 20	VIX $<$ 20	Majority-class baseline
Training (2006–2022)	36.7%	63.3%	63.3%
Test (2022–2026)	29.8%	70.2%	70.2%

High-volatility days are substantially less common in the test period ($\approx$30%) than in training ($\approx$37%), reflecting that the training window (2006–2022) encompasses several high-volatility crises—the 2008–2009 financial collapse, the 2011 eurozone episode, and the COVID-19 crash—that inflated the training-period base rate, whereas the test window (2022–2026) is dominated by the post-bear-market recovery despite the elevated volatility of the first half of 2022.

4 Methodology

4.1 Feature Engineering

The model uses 36 engineered features grouped into five categories. All features are computed using only information available at or before the prediction date t.

VIX technical indicators (13 features): 10-day and 20-day simple moving averages (SMA10, SMA20); 10-day and 20-day momentum; 10-day and 20-day rate of change; Bollinger Band standard deviation, upper and lower bands, and position within the band; 14-day RSI; moving-average cross (SMA10 $-$ SMA20); and VIX term spread (VIX3M $-$ VIX).

SKEW features (3 features): SKEW level, 5-day change, and 20-day change.

S&P 500 features (7 features): 1-day, 5-day, and 20-day returns; 10-day and 20-day price momentum; 20-day realized volatility; and 252-day maximum drawdown.

Cross-asset features (9 features): Gold 1-day, 5-day, and 20-day returns; 10-Year Treasury Yield 1-day, 5-day, and 20-day changes; DXY 1-day, 5-day, and 20-day returns.

Macroeconomic features (4 features): Federal Funds Rate 1-month and 3-month changes; yield curve spread 1-month and 3-month changes.

4.2 Models and Training Pipeline

Two ensemble classifiers are evaluated: Random Forest (RF) and Histogram Gradient Boosting (HGB), implemented via scikit-learn's `HistGradientBoostingClassifier`. This implementation is distinct from the XGBoost library [Chen and Guestrin, 2016]; it uses a histogram-binning approach that is computationally efficient for large feature sets. **Both models use a single fixed specification across all tables:** RF with `n_estimators=200, max_depth=10, min_samples_leaf=5, random_state=42`; HGB with `max_iter=200, max_depth=5, learning_rate=0.1, random_state=42`. Using one specification eliminates pipeline inconsistencies and provides a fair comparison across tables.

All features are standardized using `StandardScaler` fit on the training set and applied to the test set.

4.3 Probability Calibration

The selective prediction rule requires well-calibrated probability estimates. We apply isotonic regression calibration using `CalibratedClassifierCV` with 5-fold `TimeSeriesSplit` cross-validation on the full training set. The calibrated probability at time t is denoted $\hat{p}_t$, representing the estimated probability that VIX will be in a high-volatility regime at $t + N$.

4.4 Confidence-Based Selective Prediction

The selective classifier makes a prediction only when $\hat{p}_t$ is sufficiently far from 0.5. For confidence threshold $\tau \in (0, 0.5)$, the prediction rule is:

$$\hat{y}_t = \begin{cases} 1 & \text{if } \hat{p}_t > 0.5 + \tau \\ 0 & \text{if } \hat{p}_t < 0.5 - \tau \\ \text{abstain} & \text{if } 0.5 - \tau \leq \hat{p}_t \leq 0.5 + \tau \end{cases} \tag{3}$$

Accuracy is evaluated only on days where the model makes a prediction. Coverage is defined as the fraction of test days on which a prediction is made. The primary threshold is $\tau = 0.25$.

4.5 Diagnostic Protocol Design

The six diagnostic components are not an arbitrary collection; each targets a specific failure mode that aggregate accuracy cannot detect, and together they form a sufficient battery for distinguishing genuine predictive skill from persistence exploitation.

(i) Coverage-matched baseline comparison tests the null hypothesis that the observed accuracy gain is attributable to selective coverage of easy-to-predict regime-stable days rather than to multi-feature learning. A selective classifier that cannot outperform a coverage-matched persistence rule—which uses zero features—provides no evidence of ML skill. This test is necessary because coverage itself creates a selection effect: any classifier will look accurate if it concentrates predictions on the majority class.

(ii) Regime-transition accuracy audit identifies whether accuracy gains are localized to stable or transitional days. Transitions are the economically critical case (a risk manager needs a spike warning, not confirmation that stability persists) and are by construction the hardest for a persistent label. Zero accuracy on covered calm→high days means the model is confidently wrong on the economically relevant case—a failure invisible in aggregate accuracy.

(iii) Persistence quantification provides a mechanistic decomposition: what fraction of the model's correct covered predictions are regime-stable calls (same state at $t + N$ as at t)? A value of 99–100% means the model is empirically indistinguishable from a persistence rule on the days it predicts.

(iv) SHAP feature attribution asks what information drives predictions at the individual-prediction level. If autocorrelation proxies (VIX moving averages) account for >70% of SHAP mass and macro/cross-asset features together account for under 10%, the model is learning persistence and correlated equity dynamics rather than multi-dimensional market structure. Any residual attribution to equity or volatility features still reflects co-movement with VIX, not independent predictive content.

(v) Sequence-model comparison (LSTM) tests whether the failure is architecture-specific or information-theoretic. A pointwise classifier (RF) might fail on transitions simply because it lacks temporal context. A two-layer LSTM with 20-day sequence windows should in principle learn leading patterns in the feature trajectory before a spike. If the LSTM also fails on calm→high transitions, the constraint is in the daily feature set itself, not in the model's capacity to learn temporal dynamics.

(vi) Cross-task replication establishes external validity. If the persistence illusion appeared only in VIX regime classification, it could be attributed to the specific properties of implied volatility. Replication on S&P 500 trend regimes, yield curve inversion, and any additional autocorrelated target rules out domain-specific explanations and establishes the mechanism as structural.

Joint sufficiency. The six components form a closed diagnostic argument. Components (i)–(iii) establish *that* the illusion exists and quantify it. Component (iv) explains *what* the model learns. Component (v) establishes the explanation is *informational* (not architectural). Component (vi) establishes the mechanism is *general*. A model that passes all six—RF exceeds persistence with McNemar $b + c > 0$ on jointly covered days, calm→high accuracy above 50%, same-regime fraction $< 100\%$, SHAP dominated by non-autocorrelation features, LSTM matching RF on transitions, pattern not replicating across tasks—would constitute evidence of genuine multi-feature predictive skill. Section 6 demonstrates that the protocol identifies such a model correctly on a synthetic dataset with injected predictive signal.

5 Results

5.1 Main Accuracy Results

Table 2 reports baseline and selective accuracy, coverage, and balanced accuracy on covered days at $\tau = 0.25$, using the fixed-spec calibrated pipeline. Figure 1 plots selective accuracy, baseline accuracy, and balanced accuracy across horizons.

Table 2: Baseline (uncalibrated) and selective (calibrated, $\tau = 0.25$) accuracy, coverage, and balanced accuracy on covered days. Fixed-spec calibrated pipeline used throughout. Gain = selective minus baseline in percentage points (pp).

N	RF Base	RF Sel	RF Gain	RF Cov	RF BalAcc	HGB Sel	HGB Gain	HGB Cov	HGB BalAcc
5	83.4%	91.9%	+8.5 pp	76.3%	88.4%	92.1%	+8.9 pp	71.1%	88.2%
10	79.5%	90.0%	+10.5 pp	60.1%	84.8%	92.2%	+13.6 pp	63.1%	86.1%
15	76.2%	90.0%	+13.8 pp	51.3%	83.9%	92.2%	+21.3 pp	36.0%	89.5%
20	76.8%	92.7%	+15.9 pp	31.7%	79.1%	94.8%	+22.2 pp	26.8%	84.1%
25	73.4%	90.3%	+16.9 pp	26.9%	**50.0%**	92.7%	+26.5 pp	19.2%	**50.0%**

Table 3: Extended metrics on covered days for Random Forest ($\tau = 0.25$). At $N = 25$, precision and recall are both 0% and balanced accuracy equals 50%, confirming that the model exclusively predicts the calm class at this horizon.

N	Selective Acc	Precision	Recall	F1	Balanced Acc
5	91.9%	86.5%	81.2%	83.8%	88.4%
10	90.0%	86.8%	73.8%	79.7%	84.8%
15	90.0%	89.8%	70.8%	79.2%	83.9%
20	92.7%	100.0%	58.2%	73.6%	79.1%
25	90.3%	0.0%	0.0%	0.0%	**50.0%**

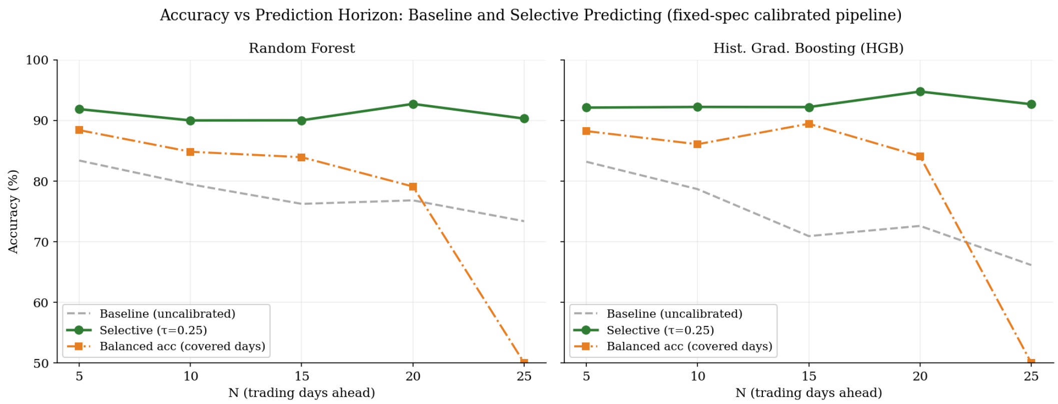

Figure 1: Baseline, selective accuracy, and balanced accuracy across prediction horizons for Random Forest (left) and HGB (right). Selective accuracy (green) remains high across horizons while baseline accuracy (gray dashed) declines. Balanced accuracy (orange) reveals the class-imbalance artifact: at $N = 25$, balanced accuracy equals 50%, confirming a coin-flip result despite 90% selective accuracy.

Selective prediction improves accuracy for both models at every horizon. However, the balanced accuracy columns tell a different story: RF balanced accuracy declines monotonically from 88.4% at $N = 5$ to 79.1% at $N = 20$, and collapses to 50.0% at $N = 25$. This 50% balanced accuracy at $N = 25$ confirms that the model exclusively predicts low-volatility outcomes on covered days at the longest horizon—a result confirmed by the extended metrics in Table 3, where precision and recall are both 0% for the positive class. HGB exhibits the same degeneracy (HGB BalAcc = 50.0% at $N = 25$). The nominal 90%–92% selective accuracy at $N = 25$ for both models reflects the 70% base rate of calm days, not predictive skill. As the balanced accuracy columns make clear, $N \leq 20$ represents the practically informative prediction window.

At $N = 20$, precision reaches 100% (no false positive spike warnings on covered days) but recall falls to 58% (42% of actual spikes missed). This precision-recall tradeoff is characteristic of a model that abstains on uncertain days: the covered days are predominantly clear-cut, while the borderline transition days (most likely to be spikes) fall in the abstention zone.

The accuracy-coverage tradeoff is monotone: across $\tau \in \{0.10, \ldots, 0.40\}$, selective accuracy increases with τ at all horizons, confirming that calibrated probability is a reliable ranking signal. However, increasing τ does not improve coverage of transition days—it makes confident wrong predictions on calm→high days even more selective (Section 5.4). Full threshold sensitivity results and robustness to VIX regime thresholds of 18, 20, and 22 appear in Appendices B and C. Sustained regime labels (requiring VIX ≥ 20 on all days $t + 1$ through $t + N$) are consistently easier to predict than instantaneous labels; results appear in Appendix D.

5.2 Comparison to Naive Baselines

A critical question is whether the RF selective accuracy reflects genuine multi-feature ML skill or simply VIX autocorrelation. To answer this, we construct four coverage-matched baselines plus an uncovered reference predictor, all evaluated on the same test set.

The **persistence baseline** predicts $\mathrm{VIX}_{t+N} \geq 20$ if and only if $\mathrm{VIX}_t \geq 20$, and abstains when $|\mathrm{VIX}_t - 20| < c$ for a constant c chosen to match the RF's coverage. The **LR (VIX only) baseline** applies the full calibration pipeline to a single feature (current VIX level). The **HAR-style logistic regression** uses three features motivated by the HAR model [Corsi, 2009]: current VIX (VIX_D), 5-day VIX mean (VIX_W), and 20-day VIX mean (VIX_M). The **Markov-chain baseline** estimates the 1-step transition probability matrix from training data and raises it to the Nth power to predict the probability of being in a high-volatility regime N days ahead; it represents the classical regime-switching forecast under the assumption that regimes follow a first-order Markov process, and serves as a reference for how much ML models add beyond that structural assumption. An **always-calm predictor**, which issues a calm prediction on every test day unconditionally, appears in Section 5.8 as a lower bound on cost-weighted utility. All baselines and the RF use the same 80/20 train/test split and are evaluated on identical test sets.

Table 4: RF selective accuracy vs. naive baselines at matched coverage ($\tau = 0.25$). HAR-LR = HAR-style logistic regression on (VIX_D, VIX_W, VIX_M). Bolded cell values indicate where a baseline exceeds the RF selective accuracy. Bootstrap CIs for the RF–persistence gap appear in Table 9.

N	Coverage	Persistence	LR (VIX)	**HAR-LR**	Markov	RF Sel	RF−Persist
5	76.3%	91.6%	91.8%	91.6%	83.7%	91.9%	+0.3 pp
10	60.1%	88.1%	87.6%	**91.1%**	85.3%	90.0%	+1.9 pp
15	51.3%	87.9%	88.4%	**90.2%**	84.8%	90.0%	+2.1 pp
20	31.7%	90.5%	87.9%	**94.2%**	84.5%	92.7%	+2.3 pp
25	26.9%	83.8%	86.8%	**92.0%**	82.9%	90.3%	+6.5 pp

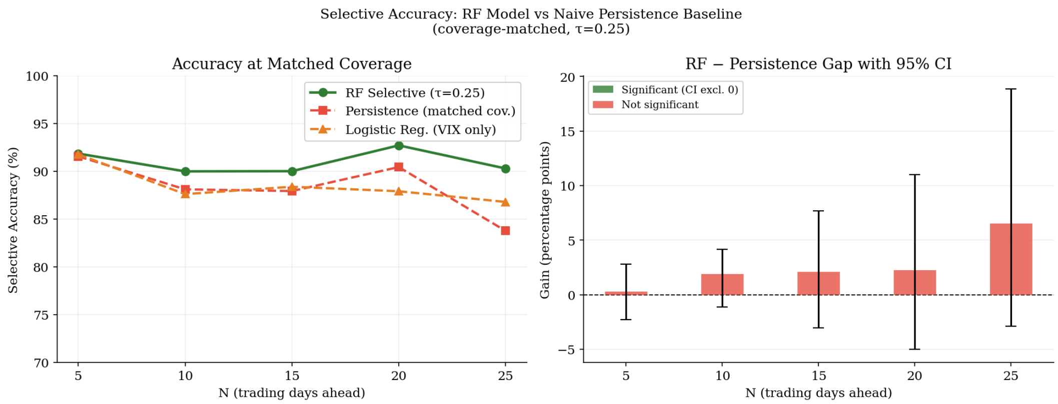

Figure 2: Left: RF selective vs. naive baseline accuracy at matched coverage across horizons. Right: RF vs. persistence gap with 95% moving-block bootstrap CI. No bar is statistically significant at 5%.

Table 4 delivers the paper's sharpest negative result. The HAR-style logistic regression—using only three VIX-related features—matches or exceeds the full 36-feature RF in point estimate at four of five horizons ($N = 10$, 15, 20, and 25). At $N = 10$, HAR-LR achieves 91.1% vs. the RF's 90.0%; at $N = 20$, 94.2% vs. 92.7%. A moving-block bootstrap on the HAR-minus-RF gap at $N = 10$ (HAR leads by +1.1 pp) yields 95% CI $[-1.08, +4.18]$; the corresponding RF-minus-persistence bootstrap gives CI $[-1.11, +4.17]$. Neither gap is statistically confirmed. The 33 additional features (equity, cross-asset, macro) contribute negligibly beyond the three-variable VIX autoregressive structure.

At $N = 5$, the RF selective (91.9%), HAR-LR (91.6%), LR-VIX (91.8%), and persistence (91.6%) are all essentially identical, confirming that the short-horizon signal is entirely VIX autocorrelation. Notably, the single-feature LR-VIX (91.8%) marginally outperforms the three-feature HAR-LR (91.6%) at this horizon, suggesting that adding the weekly and monthly VIX means provides no benefit at $N = 5$.

The RF–persistence gap (+0.3 to +6.5 pp) is directionally consistent but as Section 5.6 shows, none reaches statistical significance. The $N = 25$ gap (+6.5 pp) should be discounted because the RF at this horizon exclusively predicts calm outcomes (Table 3), inflating its nominal accuracy via the class-imbalance base rate; HAR-LR at $N = 25$ (92.0%) achieves a higher figure, though its behavior at this degenerate horizon is similarly difficult to interpret without extended metrics.

5.3 Persistence Quantification

A natural question is how much of the RF's covered accuracy is attributable to simply predicting "same regime as today." We quantify this directly. For each horizon N, we compute the empirical transition probabilities from the full dataset: $\Pr(\text{VIX}_{t+N} \geq 20 \mid \text{VIX}_t \geq 20)$ and $\Pr(\text{VIX}_{t+N} \geq 20 \mid \text{VIX}_t < 20)$. We then classify each correct RF covered prediction as either a same-regime call (predicted regime matches $\text{VIX}_t \geq 20$) or a regime-change call.

Table 5: Empirical VIX transition probabilities and decomposition of RF correct covered predictions. $\text{P(H|H)} = \text{P}(\text{VIX}_{t+N} \geq 20 \mid \text{VIX}_t \geq 20)$; $\text{P(H|C)} = \text{P}(\text{VIX}_{t+N} \geq 20 \mid \text{VIX}_t < 20)$. "Same-regime %" = fraction of correct RF covered predictions that match the regime at t.

N	P(H|H)	P(H|C)	Persist. ratio	RF cov. acc	Same-regime %
5	84.2%	8.6%	9.8×	91.9%	99.0%
10	79.5%	11.2%	7.1×	90.0%	99.6%
15	75.9%	13.2%	5.8×	90.0%	99.8%
20	73.1%	14.8%	4.9×	92.7%	100.0%
25	70.0%	16.5%	4.2×	90.3%	100.0%

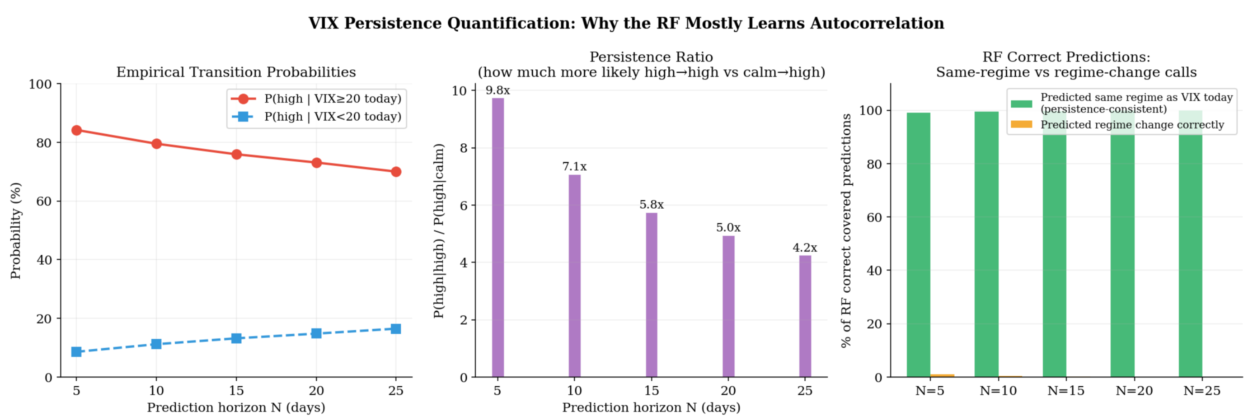

Figure 3: Left: empirical transition probabilities $\text{P}(\text{VIX}_{t+N} \geq 20)$ conditional on today's regime. Center: persistence ratio P(H|H)/P(H|C), quantifying how much stronger the autocorrelation signal is for current-high vs. current-calm days. Right: decomposition of RF correct covered predictions into same-regime and regime-change calls. At $N \geq 20$, 100% of correct predictions are same-regime.

At $N = 5$, the probability of VIX being high in 5 days is 84.2% given it is high today, compared to only 8.6% given it is calm today—a 9.8× ratio. This autocorrelation declines with horizon but remains 4.2× at $N = 25$. On covered days, 99.0% of the RF's correct predictions at $N = 5$ are same-regime calls; at $N \geq 20$, every correct covered prediction is a persistence call. The RF adds no directional forecasting beyond VIX autocorrelation on the days it chooses to cover.

5.4 Regime Transition Analysis

The selective prediction rule abstains when the model is uncertain; understanding *which days* it covers reveals the economic content of the confidence filter. We classify each test day by the regime at time t and the realized regime at $t + N$, creating four transition categories: calm→calm, calm→high, high→calm, and high→high.

Table 6: RF selective coverage and accuracy by regime transition type ($\tau = 0.25$, test set). Coverage = fraction of days in that transition category where the model makes a prediction. Accuracy = correctness on covered days only. Calm→high accuracy is **0%**: every covered prediction on a day about to spike is a confident "stay calm" signal.

Transition	$N = 5$			$N = 10$		
	Days	Coverage	Acc (covered)	Days	Coverage	Acc (covered)
calm → calm	614	87.1%	99.6%	587	71.6%	100.0%
calm → high	79	45.6%	**0.0%**	101	41.6%	**0.0%**
high → calm	84	35.7%	23.3%	111	18.0%	10.0%
high → high	222	72.5%	99.4%	199	59.3%	100.0%

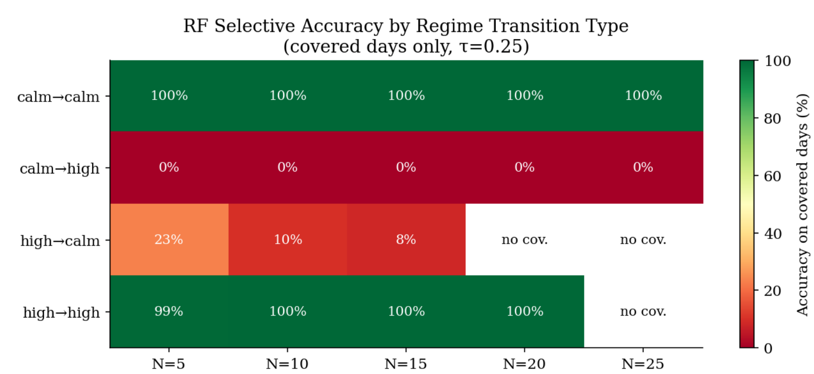

Figure 4: RF selective accuracy by regime transition type and horizon (covered days only, $\tau = 0.25$). Green = high accuracy; red = low. Calm→high rows are uniformly red at all horizons.

The table reveals the fundamental limitation of the framework. At $N = 5$, the model covers 36 of 79 calm→high days (45.6%). On *every single one* of those 36 days, it issues a confident "stay calm" prediction—and is wrong. The mechanism is clear: VIX comfortably below 20 generates calibrated probabilities below 0.25, triggering a high-confidence calm prediction. The model is correct that such days usually remain calm, but on the minority of cases where a spike follows, it is confidently mistaken.

Structural caveat. This failure is not a model-specific defect: it is an expected property of the prediction problem. Calm→high transitions are disproportionately triggered by exogenous events—geopolitical shocks, central bank surprises, sudden liquidity dislocations, pandemic-scale disruptions—whose onset leaves no advance signature in publicly available daily price and macro features. A model built on such features is analogous to a thermometer used as an earthquake predictor: it measures something real, performs well under normal conditions, but is structurally blind to the shock class because the information simply is not in the instrument. The diagnostic value of 0% covered accuracy is therefore not that the model is badly calibrated, but that the confidence filter fails to surface its own ignorance—issuing confident wrong predictions on covered transition days rather than abstaining. An early-warning application would require either features that carry event-anticipation content (options skew, credit spreads, order-flow imbalance) or a fundamentally different modeling target.

The aggregate accuracy figures (89–92%) are therefore driven almost entirely by the two diagonal cells (calm→calm and high→high), which together account for over 80% of test days. The off-diagonal cells both show poor accuracy: calm→high covered accuracy is 0%, while high→calm covered accuracy is 23.3% at $N = 5$ and 10.0% at $N = 10$—both substantially below the 50% random baseline. On days when VIX is elevated but will revert to calm within N days, the model confidently predicts continued high volatility and is wrong more often than chance. A practitioner using this framework as a spike early-warning system would receive confident, wrong "all clear" signals on nearly half of the days before a spike materialized.

Event case studies. Figure 5 documents the model's behavior during four major volatility events in the test set: the 2022 bear-market VIX peak (June 2022), the UK gilt crisis (October 2022), the SVB banking collapse (March 2023), and the Japan carry-trade unwind (August 2024, when VIX reached 65). Each panel shows the calibrated probability $\hat{p}_t$ for the 40 trading days leading up to the spike, with predictions classified as confident calm, confident high, or abstaining.

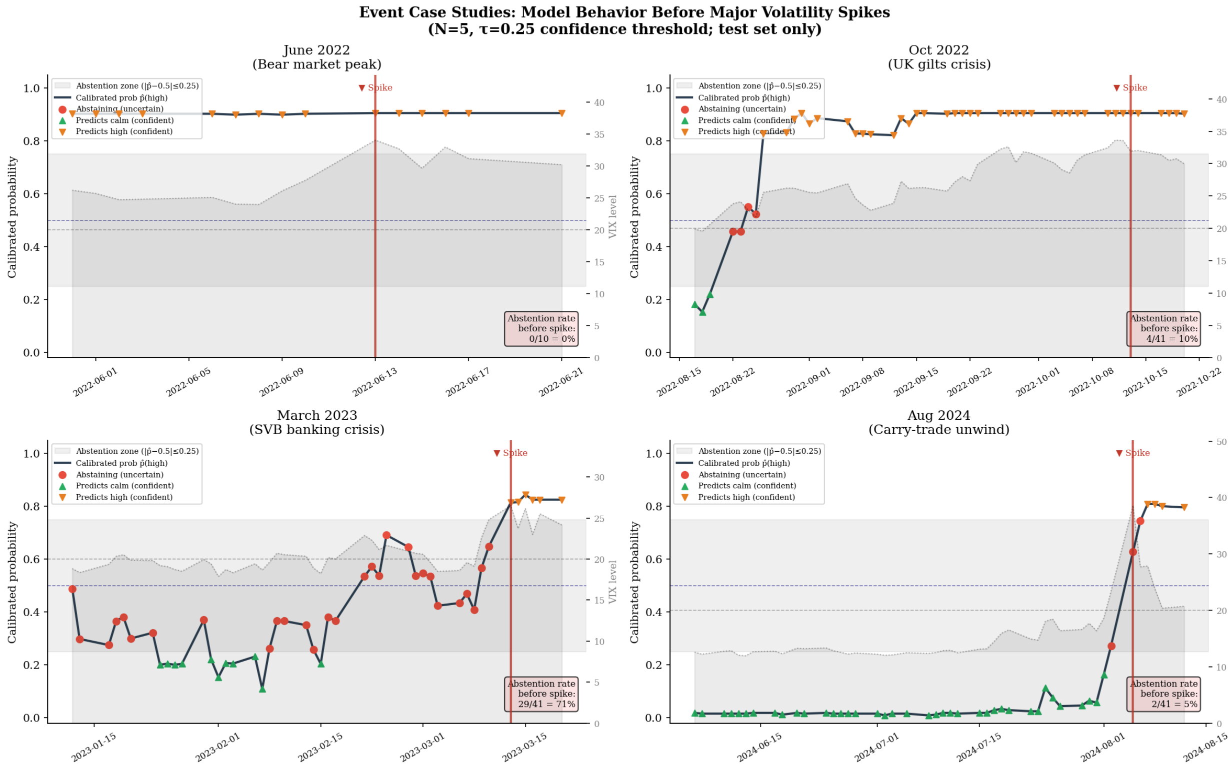

Figure 5: Model behavior ($\hat{p}_t$, $N = 5$, $\tau = 0.25$) for 40 trading days before four major volatility events in the test period. Red circles = abstaining; green triangles = confident calm; orange inverted triangles = confident high. The gray band is the abstention zone ($|\hat{p}_t - 0.5| \le 0.25$). VIX level is shown as a dotted background line; the vertical red marker is the spike date. In all four cases the model issues confident calm predictions leading up to the event.

Across all four events the pattern is consistent: the model issues confident "stay calm" predictions on the days leading up to each spike and does not show elevated abstention rates before the event. The August 2024 carry-trade unwind—the most dramatic test-period spike (VIX $\approx$ 65)—shows essentially no anticipatory signal; $\hat{p}_t$ remains below $0.5 - \tau$ until the day the spike materializes. These case studies translate the aggregate 0% calm$\to$high covered accuracy into concrete practitioner experience: a risk manager relying on this framework would have received confident all-clear signals on the days before each of these crises.

Cost-weighting does not rescue calm→high recall. To test whether the 0% calm$\to$high covered accuracy reflects an objective-function deficiency rather than an information ceiling, we refit the RF with explicit class weights $\{0{:}1, 1{:}w\}$ for $w \in \{1, 5, 10, 20\}$, penalizing false negatives on high-volatility days by up to 20× relative to false positives. Results appear in Table 7. At $N = 5$, calm→high day coverage increases modestly from 45.6% to 49.4% as w rises, but covered accuracy remains **0.0% at every weight**. At $N = 10$, calm→high coverage increases more substantially from 41.6% to 58.4% under $w = 10$, yet every covered prediction is still a confident "stay calm." The coverage increase under heavy weighting is not paradoxical: aggressive class-weighting pushes the RF to assign higher probabilities to the days it classifies as high-vol; the subsequent isotonic calibration then maps the remaining days—including calm→high transitions, which are feature-indistinguishable from ordinary calm days—to even lower calibrated probabilities, pushing more of them below $0.5 - \tau$ and into the high-confidence calm zone rather than the abstention zone. Cost-weighted utility also worsens monotonically with w: at $N = 10$, the unweighted RF achieves utility -0.730 under cost ratio $C = 10$ (Table 10), and increasing the class weight strictly worsens this figure at every cost ratio tested. The class-weighting experiment delivers a clean verdict: the failure to predict calm→high transitions is *informational*, not objective-function-driven. The daily feature set does not contain enough advance information about shock onset to produce correct predictions on transition days regardless of how heavily the training objective penalizes misses.

Table 7: Effect of class-weighting on transition coverage and selective accuracy. w = false-negative class weight; $\tau = 0.25$ throughout. C→H = calm→high; C→C = calm→calm. Calm→high accuracy is **0.0%** at every weight for both horizons.

N	w	C→H Cov	C→H Acc	C→C Cov	C→C Acc	Sel Acc
5	1	45.6%	**0.0%**	87.1%	99.6%	91.9%
5	5	46.8%	**0.0%**	87.1%	100.0%	92.2%
5	10	49.4%	**0.0%**	87.3%	100.0%	92.3%
5	20	48.1%	**0.0%**	84.4%	100.0%	91.8%
10	1	41.6%	**0.0%**	71.6%	100.0%	90.0%
10	5	49.5%	**0.0%**	77.7%	100.0%	89.5%
10	10	58.4%	**0.0%**	80.4%	100.0%	87.4%
10	20	57.4%	**0.0%**	82.8%	100.0%	87.4%

Abstention as a leading indicator. The transition analysis reveals a subtler and more positive finding. Although the model cannot make correct *predictions* on calm→high days, we test whether its *abstention rate* is elevated in the days leading up to such transitions.

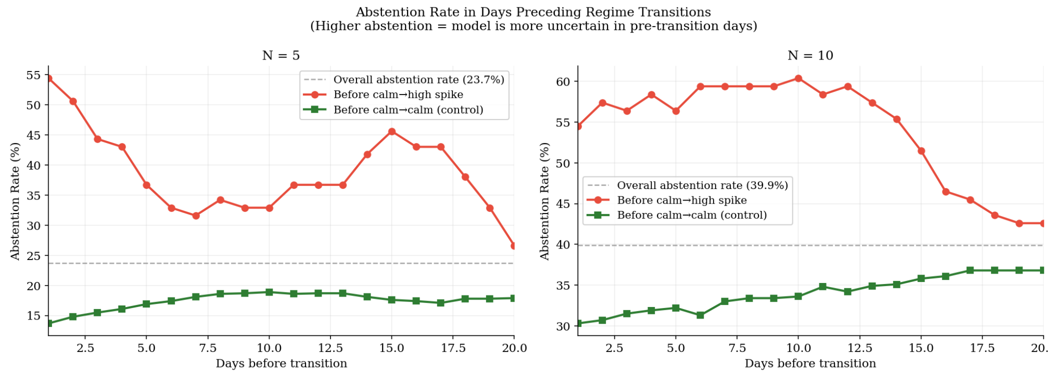

Figure 6: Abstention rate in the days preceding calm→high transitions (red) vs. calm→calm days (green), compared to the overall abstention rate (dashed). At lag = 1 day, the abstention rate before spikes is more than double the overall rate.

Figure 6 shows that at $N = 5$, the abstention rate in the day immediately before a calm→high transition is 54.4%, compared to 13.7% before calm→calm days and 23.7% overall—a 2.3× elevation. At $N = 10$, the corresponding rates are 54.5% vs. 30.3% vs. 39.9% (1.4×). This observation carries a necessary qualification: it is a post-hoc pattern measured on the same held-out test set used for all other results, and should be treated as hypothesis-generating rather than confirmatory.

Confound test: abstention vs. VIX proximity. A natural concern is that the abstention elevation trivially reflects VIX proximity to the threshold: calm→high transitions start from $\mathrm{VIX}_t < 20$ by definition, and days near the boundary may generate uncertain probabilities simply because the feature distribution near $\mathrm{VIX}_t \approx 20$ is ambiguous. To test this, we compare three signals as binary spike classifiers on calm days only ($\mathrm{VIX}_t < 20$): the raw abstention indicator, the proximity score $1 - (\mathrm{VIX}_t^{\max} - \mathrm{VIX}_t)/(\mathrm{VIX}_t^{\max} - \mathrm{VIX}_t^{\min})$ where the range is taken over calm-day test observations,[1] and the HAR-LR calibrated probability. We evaluate each with ROC/AUC and apply a moving-block bootstrap (2,000 replicates, block length ≈ 32, matching Section 5.6) to the abstention-rate gap.

[1]An equivalent range-free formulation is $20 - \mathrm{VIX}_t$ (negated distance), which yields the same ranking without depending on the test-set min/max. We use the normalized form for AUC comparability with the probability scores.

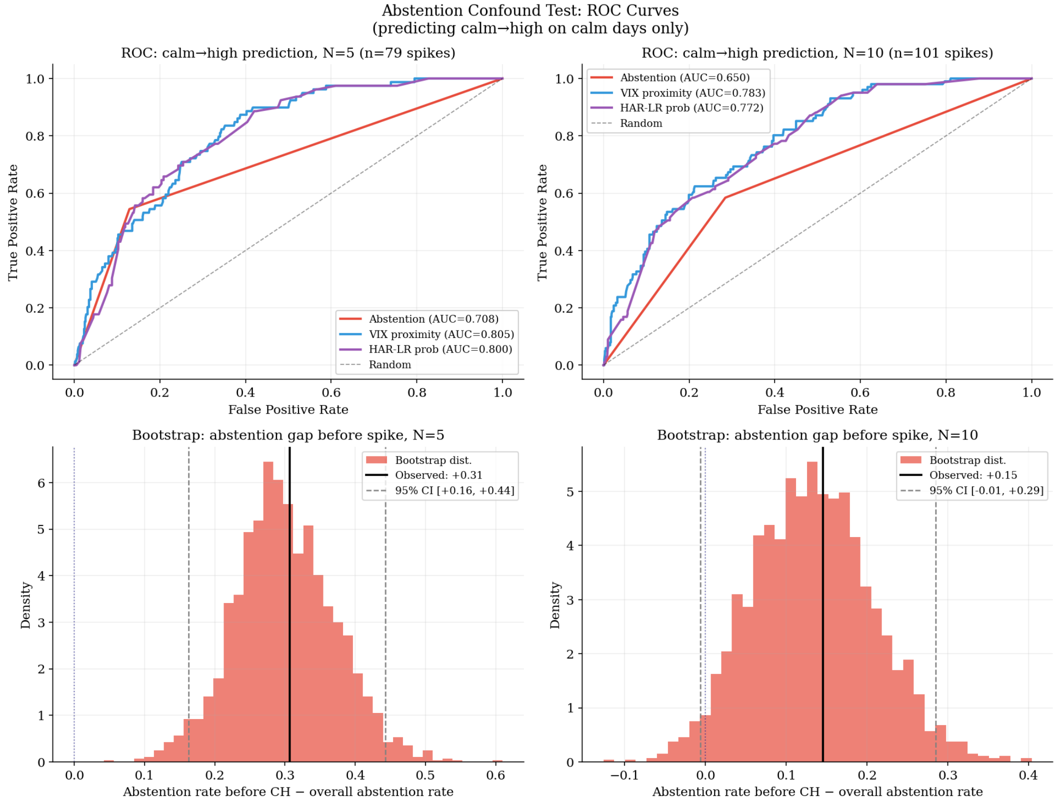

Figure 7: Top: ROC curves comparing abstention, VIX proximity to threshold, and HAR-LR probability as calm→high predictors on calm days ($\mathrm{VIX}_t < 20$), for $N = 5$ (left) and $N = 10$ (right). Bottom: Moving-block bootstrap distribution of the abstention-rate gap (before calm→high events minus overall rate), with 95% CI.

Figure 7 (top panels) shows that VIX proximity (AUC $= 0.805$ at $N = 5$; 0.783 at $N = 10$) and HAR-LR probability (AUC $= 0.800$; 0.772) both outperform raw abstention (AUC $= 0.708$; 0.650) as spike classifiers on calm days. The majority of the abstention signal is therefore accounted for by VIX proximity to the threshold.

To test whether abstention carries *incremental* information beyond proximity, we fit two logistic regressions on calm days: one using proximity alone and one using proximity plus the abstention indicator. At $N = 5$, the combined model (AUC $= 0.816$) beats proximity alone (AUC $= 0.805$); the likelihood-ratio statistic on the evaluation data corresponds to $p = 0.001$. However, because the test set was used for all other evaluations in this paper and was not reserved for this test, the p-value does not constitute a valid significance claim—it is a descriptive measure of fit improvement on this sample, not a confirmatory finding, and should be treated accordingly. At $N = 10$, the combined model adds nothing (AUC $= 0.782$ vs. 0.783, $p = 1.000$). We treat the $N = 5$ increment as a provisional regularity warranting dedicated out-of-sample confirmation. Abstention therefore carries a small residual signal beyond threshold proximity at the shorter horizon only.

The bootstrap confidence intervals on the abstention-rate gap are: $N = 5$: $+0.307$, 95% CI $[+0.162, +0.443]$ (significant); $N = 10$: $+0.146$, 95% CI $[-0.006, +0.285]$ (marginally insignificant). The overall picture is that proximity explains most of the elevation, but at $N = 5$ the model's uncertainty carries a small independent informational increment. We report this as a documented regularity warranting follow-up investigation, not as a robust predictive finding.

Five-fold expanding walk-forward validation (Appendix E) confirms that persistence at matched coverage equals or exceeds RF selective accuracy at all horizons ($N \in \{5, 10, 20\}$). The RF's one measurable advantage over persistence at $N = 5$ is fold-to-fold stability (std 1.7% vs. 3.2%), not directional accuracy. At $N = 20$, both RF and HGB collapse to fold standard deviations exceeding 25 pp; the practical reliable prediction window is $N \leq 10$.

5.5 LSTM Comparison

A natural question is whether a sequence-aware neural network—which can in principle learn from the temporal ordering of past market states—overcomes the transition-blindness identified in Section 5.4. We train a two-layer Long Short-Term Memory (LSTM) network on sliding windows of the past 20 trading days of all 36 features, using the same 80/20 chronological split. The LSTM uses hidden size 64, dropout 0.3, and is trained for 40 epochs with the Adam optimizer; calibration is applied via isotonic regression on a held-out portion of the training set. Table 8 and Figure 8 compare LSTM and RF at $\tau = 0.25$.

Table 8: LSTM vs. RF at $\tau = 0.25$ selective threshold. RF values are computed on the same trimmed test sample as the LSTM (first 20 observations consumed by the sequence window). BalAcc = balanced accuracy on covered days.

N	LSTM Sel Acc	LSTM Cov	LSTM BalAcc	RF Sel Acc	RF Cov
5	76.5%	87.0%	76.0%	91.7%	77.0%
10	71.8%	90.7%	64.7%	89.3%	64.3%

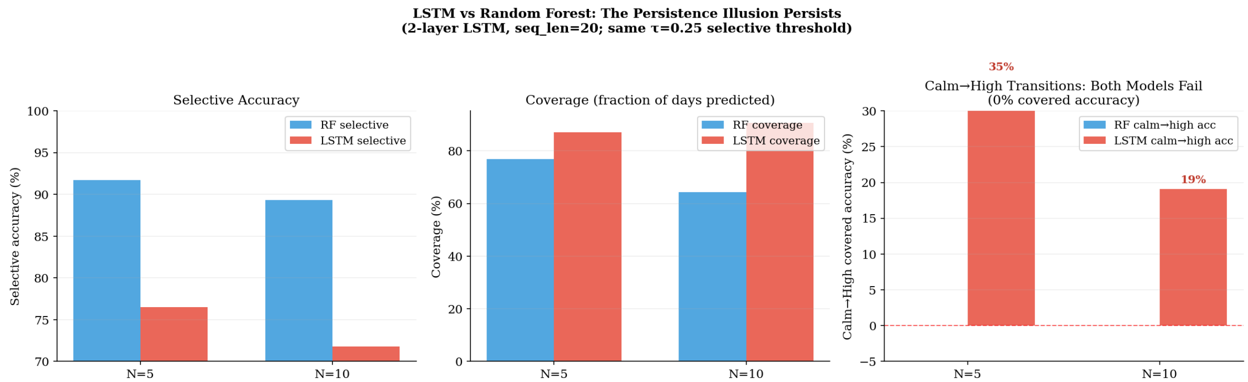

Figure 8: LSTM vs. RF comparison at $\tau = 0.25$. Left: selective accuracy. Center: coverage fraction. Right: calm→high covered accuracy for both models—the RF achieves 0% (refuses to cover) while the LSTM achieves 19–35% (covers but is wrong more often than chance). Neither model provides reliable transition forecasting.

The LSTM achieves lower selective accuracy than the RF at both horizons. One structural caveat: the LSTM operates at higher coverage than the RF at the same τ (87.0% vs. 77.0% at $N = 5$; 90.7% vs. 64.3% at $N = 10$), so the raw selective accuracy comparison is not coverage-matched. The lower LSTM accuracy reflects both lower precision on covered days and broader coverage of uncertain days—it is not a pure like-for-like comparison. The relevant diagnostic use is the transition audit, not the aggregate accuracy ranking. On the regime-transition audit, the LSTM shows non-zero calm→high covered accuracy (34.9% at $N = 5$; 19.1% at $N = 10$)—unlike the RF's 0%—because the LSTM's calibrated probabilities are less tightly clustered near the prediction boundaries, so it makes predictions on more uncertain days including some transition days. However, 34.9% and 19.1% are both substantially below the 50% random-chance baseline, meaning the LSTM is confidently wrong more often than right when it does predict on transition days. This broader coverage of transition days comes entirely at the expense of precision on stable days: calm→calm covered accuracy is 83.1% at $N = 5$ compared to the RF's 99.6%. The LSTM's overall lower accuracy reflects this tradeoff.

The result is therefore complementary to the RF finding rather than contradicting it. The two models fail differently on calm→high transitions—the RF abstains or confidently predicts calm; the LSTM predicts but is wrong more often than chance—but neither extracts reliable advance signal about regime transitions from the available daily feature set. Architectural richness (20-day temporal memory vs. pointwise features) does not resolve the information constraint identified in

Section 5.4: the daily price and macro feature set does not carry the event-anticipation content needed to forecast volatility spikes.

Calibration reliability diagrams for the RF (not the LSTM) appear in Appendix F; RF Brier scores are 0.112 at $N = 5$ and 0.142 at $N = 10$, both well below the no-skill baseline of $\bar{p}(1-\bar{p}) \approx 0.21$, confirming that the RF's calibrated probabilities are well-ordered and the abstention criterion is applied to meaningful probability estimates.

5.6 Statistical Significance of the Persistence Gap

Section 5.2 reports an RF vs. persistence accuracy gap of +1.9 pp at $N = 10$ and +2.1–2.3 pp at $N = 15$–20. To assess statistical significance under serial dependence, we apply a moving-block bootstrap to the paired accuracy difference (RF selective minus persistence, at matched coverage). Block length is set to $\max(\sqrt{N_{\text{test}}}, 20) \approx 32$ days following Kunsch [1989]; 2,000 bootstrap replicates are drawn; CI endpoints vary by less than 0.1 pp across random seeds.

Table 9: Moving-block bootstrap 95% confidence intervals for the RF vs. persistence accuracy gap at matched coverage ($\tau = 0.25$). All CIs include 0; no gap is statistically significant at the 5% level.

N	Gap (pp)	95% CI lower	95% CI upper
5	+0.30	−2.26	+2.80
10	+1.88	−1.11	+4.17
15	+2.07	−3.03	+7.70
20	+2.28	−5.01	+11.03
25	+6.52	−2.86	+18.87

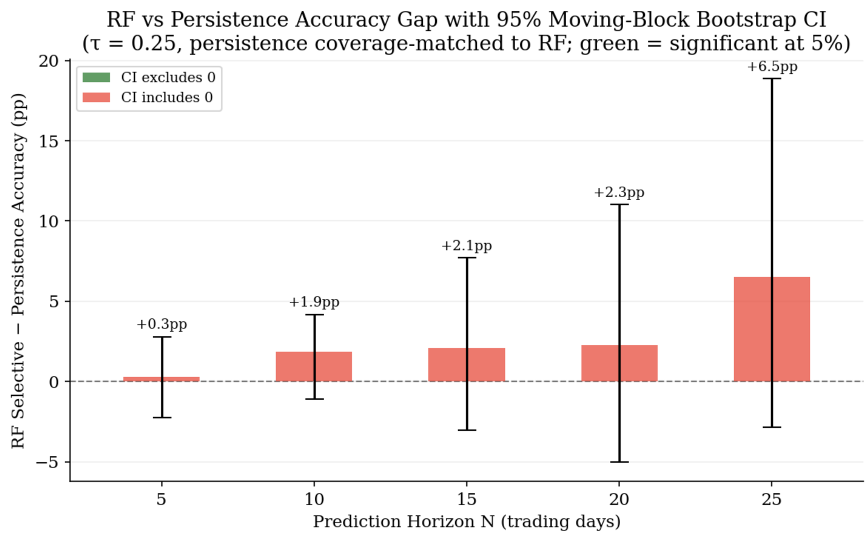

Figure 9: RF vs. persistence accuracy gap with 95% moving-block bootstrap CI. No bar is statistically significant at 5%; all CIs include 0. $N = 10$ is closest: +1.88 pp, CI $[-1.11, +4.17]$.

No gap is statistically significant at the 5% level; all confidence intervals include 0. The $N = 10$ result (+1.88 pp, 95% CI $[-1.11, +4.17]$) is the closest to significance. However, the McNemar test below provides a sharper interpretation: because RF and persistence make identical predictions on every jointly covered day at $N \geq 10$, the gaps in Table 9 are attributable entirely to differences in *which* days each rule covers, not to differing predictions on shared days. The open question is therefore whether the RF's coverage *selection*—not its predictions—carries independent value; the walk-forward stability at $N = 5$ (RF fold std 1.7% vs. persistence 3.2%) weakly suggests it might, but the directional accuracy gap does not indicate latent multi-feature predictive skill awaiting more data.

McNemar's test: RF vs. persistence on jointly covered days. A paired McNemar's test on jointly covered days—days where both the RF selective rule and the coverage-matched persistence rule make a prediction—establishes this composition interpretation precisely. We use the mid-p McNemar variant, which is exact for small $b + c$ and coincides with the chi-squared form when $b + c \geq 25$. At $N = 10$, 15, 20, and 25, the contingency table has $b = c = 0$: RF and persistence make *identical* predictions on every jointly covered day ($\chi^2 = 0$, $p = 1.000$ at all four horizons). At $N = 5$, there are 689 jointly covered days; RF is correct where persistence is wrong on only 3 days ($c = 3$, $b = 0$, gap = +0.44 pp, $p = 0.083$). The Table 9 accuracy gaps are therefore attributable entirely to differences in which days each rule covers, not to differing predictions on shared days. At $N \geq 10$, the RF chooses to make predictions exclusively on days already far from the 20 threshold, producing the same call as the persistence rule on every one of them.

Robustness to 2022 exclusion. The year 2022 was an unusually high-volatility period that accounts for 15–17% of test days depending on horizon. Removing those 149–165 days leaves 830–850 observations per horizon. The qualitative conclusions are unchanged: RF selective still exceeds persistence by +1.25 to +7.34 pp, HAR-LR matches or beats RF at $N \geq 10$ (HAR leads by +2.24 pp at $N = 10$ and +3.43 pp at $N = 20$), and calm→high covered accuracy remains 0% at all horizons. The 2022 bear-market cluster does not explain the paper's primary findings.

5.7 SHAP Feature Attribution

SHAP (SHapley Additive exPlanations) values decompose each individual prediction into additive feature contributions grounded in cooperative game theory [Lundberg and Lee, 2017]. We compute SHAP values using `TreeExplainer` on the test set at $N = 5$ and $N = 10$; permutation importance (which avoids the high-cardinality bias of Mean Decrease in Impurity, MDI) is consistent with the SHAP ranking and appears in Appendix G.

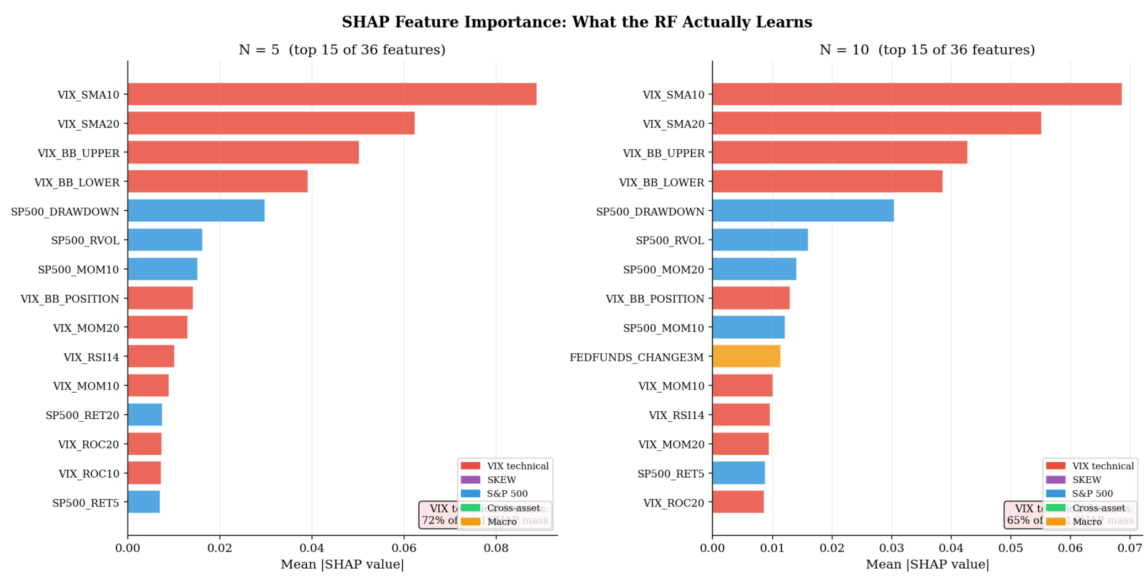

Figure 10: Mean absolute SHAP values (top 15 of 36 features) at $N = 5$ (left) and $N = 10$ (right), colored by feature category. The annotation in the lower right of each panel reports the fraction of total SHAP mass attributable to VIX technical features.

VIX technical features account for **71.9%** of total SHAP mass at $N = 5$ and **64.7%** at $N = 10$. S&P 500 features contribute a meaningful secondary share (19.2% at $N = 5$; 21.3% at $N = 10$), reflecting the well-known co-movement between equity drawdowns and volatility spikes. Macro and cross-asset features together account for under 10% at both horizons—their contribution is negligible. The dominant features—VIX_SMA10, VIX_SMA20, VIX_BB_UPPER, VIX_BB_LOWER—are all smoothed or envelope transforms of the current VIX level. SHAP therefore confirms and quantifies what the persistence analysis (Section 5.3) established: the model learns VIX autocorrelation and its equity co-movement, not independent multi-dimensional market structure. The 33 additional features beyond the three-variable HAR structure contribute 28% of SHAP mass combined, but this is concentrated in the equity category; the 20 non-VIX, non-equity features contribute under 10%.

5.8 Cost-Weighted Utility Analysis

The transition analysis shows that the framework achieves 0% accuracy on calm→high spike days it covers. From a practitioner standpoint, the asymmetry between false negatives (missed spikes) and false positives (false alarms) matters: failing to predict a volatility spike is typically far more costly than issuing an unnecessary hedge. We quantify this using a cost utility function:

$$U = -\frac{\mathrm{FP} + C \cdot \mathrm{FN}}{n_{\mathrm{predictions}}} \tag{4}$$

where C is the relative cost of a false negative vs. a false positive, and higher utility (closer to zero) is better. We compare RF selective, persistence at matched coverage, RF baseline (full coverage), and always-calm (100% coverage, zero FP, all FN) across FN/FP cost ratios $C \in \{1, 5, 10, 20\}$ at $N = 5$ and $N = 10$.

Table 10: Cost-weighted utility per prediction under different FN/FP cost ratios C. Higher utility (closer to 0) is better. Bolded entries show where persistence dominates RF selective prediction.

N	C	RF Selective	Persistence	RF Baseline	Always Calm
5	1	−0.081	−0.087	−0.166	−0.301
5	5	−0.276	−0.286	−0.515	−1.507
5	10	−0.518	−0.535	−0.950	−3.013
5	20	−1.004	−1.033	−1.821	−6.026
10	1	−0.100	−0.120	−0.205	−0.301
10	5	−0.380	**−0.326**	−0.626	−1.503
10	10	−0.730	**−0.584**	−1.152	−3.006
10	20	−1.430	**−1.100**	−2.204	−6.012

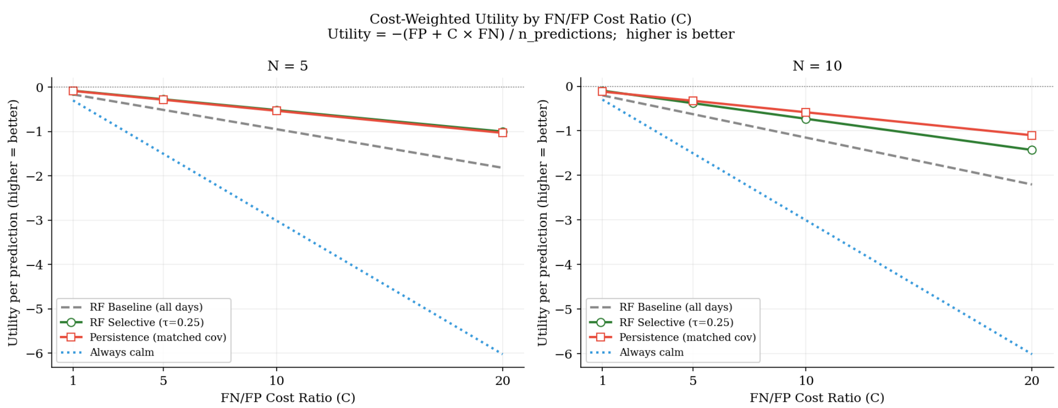

Figure 11: Cost-weighted utility as a function of FN/FP cost ratio C at $N = 5$ (left) and $N = 10$ (right). At $N = 10$ with $C \geq 5$, persistence at matched coverage dominates RF selective prediction.

At $N = 5$, RF selective beats persistence at all cost ratios, reflecting modest precision advantages at the shortest horizon. At $N = 10$, the ordering reverses when $C \geq 5$: persistence at matched coverage achieves higher utility than RF selective prediction because the persistence baseline makes fewer false negative errors on the covered days. This result quantifies the practical implication of the 0% calm→high accuracy: the RF selective framework is utility-beneficial only when false positives are as costly as false negatives ($C = 1$). At any moderate FN/FP cost asymmetry ($C \geq 5$), a practitioner would be better served by the simpler persistence rule at $N = 10$.

6 Synthetic Positive Control

Section 4.5 argues that the six diagnostic components form a battery capable of identifying genuine predictive skill as well as exposing the persistence illusion. We verify this by applying the protocol to a synthetic dataset where predictive signal for regime transitions is known to exist.

6.1 Generative Model

The generative model has two components. **Background dynamics:** a Markov chain with $P(\text{high}|\text{high}) = 0.85$ and $P(\text{high}|\text{calm}) = 0.05$—the same empirical autocorrelation as VIX at the daily horizon—produces a base regime sequence with a 23% high base rate. **Event-alarm process:** on each calm day, an independent alarm fires with probability 15% (determined by whether an exogenous event severity score $Z_t \sim \mathcal{N}(0, 1)$ exceeds the 85th percentile). When the alarm fires on a calm day, $P(\text{high in } N \text{ days} \mid \text{alarm}) = 0.90$; when it does not, $P(\text{high in } N \text{ days} \mid \text{no alarm, calm}) = 0.05$.

The feature set (36 features total) contains: five *regime-proxy* features (noisy observations of today's regime state and its rolling means and momentum—analogous to the VIX SMA, Bollinger, and momentum features in the main analysis), one *alarm-oracle* feature (a noisy observation of Z_t, which is above the alarm threshold only on alarm days), and 30 pure-noise features. The regime-proxy features allow the RF to learn persistence; the alarm-oracle allows it to detect upcoming transitions. The noise features replicate the ratio of signal to noise in the main 36-feature set.

6.2 Results

Table 11 compares the full diagnostic protocol output for the synthetic dataset and the real VIX at $N = 5$ and $N = 10$; Figure 12 visualizes the three most discriminating criteria.

Table 11: Synthetic positive-control diagnostic results vs. real VIX (from Section 5). The synthetic RF has access to a genuine alarm-oracle feature; the VIX RF does not. Protocol passes (✓) or fails (×) noted for each diagnostic criterion. **Balanced accuracy is computed on all test-set days** (covered and abstained), not on covered days only; the corresponding covered-days BalAcc for VIX RF is 88.4% ($N = 5$) and 84.8% ($N = 10$) as reported in Table 2. The all-days metric is used here because it captures the class-imbalance artifact the protocol is designed to expose.

Diagnostic criterion	Synthetic N=5	Synthetic N=10	VIX N=5	VIX N=10
Sel acc (%)	90.4	90.6	91.9	90.0
Balanced acc (%)	87.7	88.2	57.1	54.9
RF−Persist (pp)	+8.7	+7.5	+0.3	+1.9
McNemar c (RF right, persist wrong)	67	61	3	0
calm/exp→high covered acc (%)	**54.9**	**55.5**	**0.0**	**0.0**
Same-regime % of correct preds	92.1	92.8	99.0	99.6
Protocol verdict	**PASS**	**PASS**	**FAIL**	**FAIL**

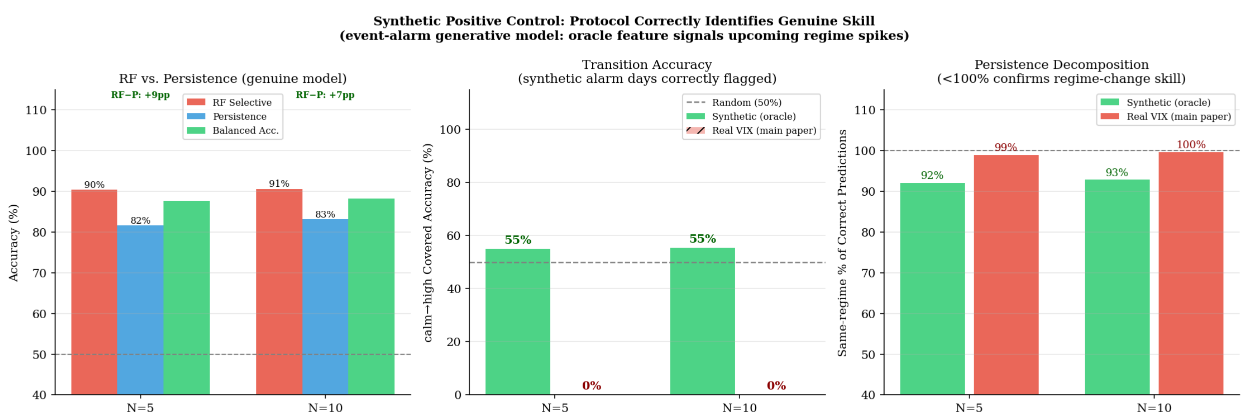

Figure 12: Synthetic positive control vs. real VIX. Left: RF beats persistence by +8.7 pp in the synthetic dataset (genuine skill), while it barely exceeds persistence in the VIX data. Center: calm→high covered accuracy is 54.9% (synthetic) vs. 0% (VIX)—the central diagnostic discriminator. Right: same-regime fraction of correct predictions is 92% (synthetic, non-trivially below 100%) vs. 99% (VIX, essentially all persistence calls).

The synthetic RF passes every diagnostic criterion where the VIX RF fails. Calm→high covered accuracy is 54.9% at $N = 5$ and 55.5% at $N = 10$ (vs. 0% for VIX at both horizons), confirming the protocol correctly identifies alarm-day transitions. RF selective accuracy exceeds matched-coverage persistence by +8.7 pp at $N = 5$ with McNemar $c = 67$: in 67 jointly covered test days, the RF is correct and persistence is wrong—genuine multi-feature skill beyond autocorrelation. The same-regime fraction of correct predictions is 92.1%, materially below the VIX value of 99%, reflecting the model's ability to make correct regime-change calls on alarm days. Balanced accuracy (87.7%) is substantially higher than the VIX model's balanced accuracy (57.1%), confirming that the genuine model is not class-imbalance-exploiting.

Notably, the alarm-oracle feature achieves only 54.9% calm→high covered accuracy even when it is genuinely informative. This reflects the irreducible fraction of calm→high transitions (roughly 40%) that occur without an alarm firing—transitions the model correctly cannot predict. The positive-control result therefore does not assert that genuine predictive skill yields 100% transition accuracy; it asserts that the diagnostic criteria (transition accuracy $> 50\%$, McNemar $c > 0$, same-regime fraction $< 100\%$) reliably distinguish a model with *some* genuine skill from one with no skill beyond persistence.

7 Replication: Cross-Task Generalizability

The positive control establishes that the protocol can identify genuine skill. The replication analysis asks whether the persistence illusion, identified in VIX regime classification, is VIX-specific or a general structural property of selective prediction on autocorrelated labels. We apply the full diagnostic protocol to two structurally different regime classification tasks using the same feature set and model specification.

7.1 Setup

Two replication tasks are evaluated. The first, described here, is **S&P 500 trend-regime classification**. The second, yield curve inversion (Section 7.3), uses a macroeconomic regime label driven by monetary policy rather than equity dynamics. Both use the same 36 features, RF specification, and $\tau = 0.25$ selective threshold as the main VIX analysis.

For S&P 500 trend-regime classification, at time t, the label is

$$\mathrm{BEAR}_{N,t} = \begin{cases} 1 & \text{if } \mathrm{SP500}_{t+N} < \overline{\mathrm{SP500}}_{200,t+N} \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

where $\overline{\text{SP500}}_{200,t+N}$ is the 200-day moving average of the S&P 500 at time $t + N$. This label captures whether the market will be in a downtrend (below its long-run average) N days ahead. It is structurally different from the VIX label—equity trend vs. implied volatility level—but shares the key property of strong autocorrelation: equity trends are persistent. The bear-market base rate in the test set (≈22%) is comparable to the VIX high-volatility base rate (≈30%). The HAR-analog baseline uses the S&P 500 level, its 5-day mean, and its 20-day mean—the equity counterpart of the VIX HAR.

7.2 Results

Tables 12 and 13 and Figure 13 present the S&P 500 replication results.

Table 12: S&P 500 trend-regime replication: RF selective vs. baselines at matched coverage ($\tau = 0.25$). RF−Persist is *negative* at every horizon: persistence at matched coverage strictly dominates the RF selective classifier. HAR-analog = logistic regression on (SP500_t, SP500_W, SP500_M).

N	Base rate	Coverage	Persistence	HAR-analog	RF Sel	RF−Persist
5	22.0%	74.4%	**100.0%**	86.9%	96.0%	−4.0 pp
10	21.9%	68.3%	**99.6%**	86.4%	94.9%	−4.7 pp
15	21.9%	69.7%	**97.3%**	86.7%	90.9%	−6.3 pp
20	21.8%	70.3%	**94.9%**	84.9%	87.7%	−7.1 pp
25	21.7%	59.5%	**94.8%**	84.8%	88.3%	−6.4 pp

Table 13: S&P 500 transition accuracy and persistence quantification. Bull→bear covered accuracy is **0%**—exact mirror of VIX calm→high. Same-regime % = fraction of correct RF covered predictions that are persistence-consistent calls.

N	Transition	Days	Coverage	Acc (covered)	Same-regime %
5	bull→bull	749	88.8%	100.0%	99.2%
5	bull→bear	25	68.0%	**0.0%**	
10	bull→bull	730	86.2%	100.0%	98.6%
10	bull→bear	39	71.8%	**0.0%**	

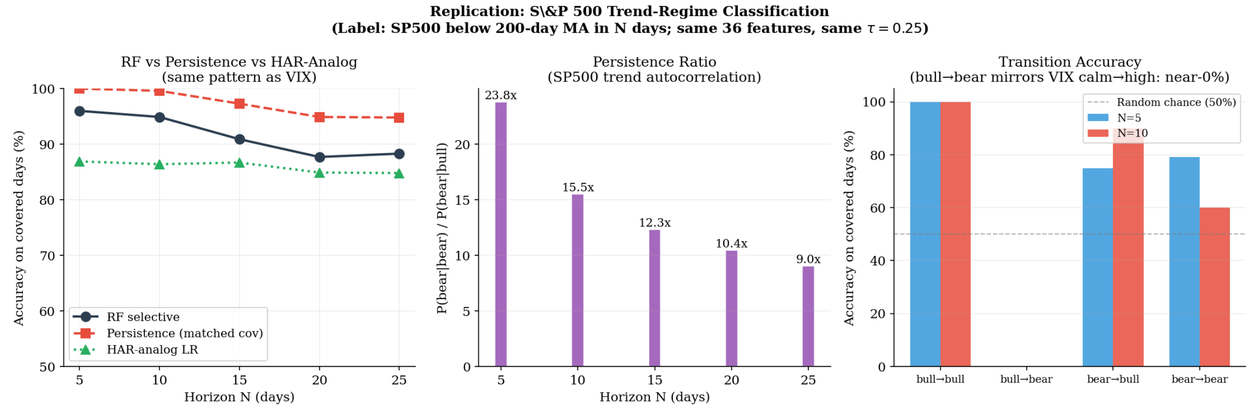

Figure 13: S&P 500 trend-regime replication. Left: RF vs. persistence vs. HAR-analog accuracy—persistence dominates at all horizons. Center: persistence ratio P(bear|bear)/P(bear|bull), which is 23.8× at $N = 5$ (higher than VIX's 9.8×). Right: transition accuracy—bull→bear covered accuracy is 0% at both horizons, replicating the VIX calm→high finding exactly.

The replication results match the VIX findings in every qualitative respect and strengthen them quantitatively. The persistence ratio is **23.8**× at $N = 5$—more than twice the VIX ratio of 9.8×—reflecting the well-known long memory of equity trends. As a result, persistence at matched coverage achieves 100% accuracy at $N = 5$ and beats the RF selective classifier at *every horizon* (RF−Persist ranges from −4.0 to −7.1 pp). Bull→bear covered accuracy is **0.0%** at both $N = 5$ and $N = 10$—exactly replicating the VIX calm→high finding—and 97–99% of the RF's correct covered predictions are same-regime calls. Balanced accuracy collapses toward 50% at $N = 25$ (extended metrics not tabulated), mirroring the VIX degeneration.

The one quantitative difference is that the RF selective accuracy (≈88–96%) appears high, but this is explained entirely by the ≈78% majority-class bull rate on covered days combined with 100% bull→bull accuracy. The model is a perfect persistence machine on the covered days it selects, and those days are overwhelmingly regime-stable.

7.3 Yield Curve Inversion Regime

The S&P 500 replication uses an equity-market label. We include a second replication on a **macroeconomic** regime: yield curve inversion, defined as $\text{T10Y2Y}_{t+N} < 0$. This label is structurally distinct from both VIX (implied volatility level, equity-driven) and S&P 500 trend (equity price momentum): it is driven by monetary policy and rate dynamics, not equity markets. The test period (June 2022–May 2026) includes the 2022–2023 inversion episode when T10Y2Y fell below

−1.0%, giving genuine class variation. The test base rate is approximately 54% inverted days.

One methodological note: the 36 features include only T10Y2Y *changes* (1-month and 3-month; features 35–36), not the current spread level. The persistence baseline, by contrast, uses the current T10Y2Y level directly ($\text{T10Y2Y}_t < 0$). This means the model must infer the current inversion state from changes alone, while the persistence rule observes it directly—a structural asymmetry that makes this a conservative test from the model's perspective. The purpose of including this replication is not to evaluate an optimally specified yield curve model, but to test whether the persistence illusion mechanism reappears when the persistence ratio is substantially higher than in VIX or S&P 500.

Table 14: Yield curve inversion replication: RF selective vs. persistence at matched coverage. RF−Persist is negative at every horizon. Persistence ratio at $N = 5$ is 75.5×—far exceeding VIX (9.8×) and S&P 500 (23.8×)—reflecting the multi-month duration of yield curve inversions.

N	Base rate	Coverage	Persistence	RF Sel	RF−Persist	Persist ratio
5	54.4%	100.0%	98.8%	45.8%	−53.0 pp	75.5×
10	54.6%	100.0%	98.0%	45.8%	−52.2 pp	44.9×
15	54.9%	100.0%	97.0%	45.7%	−51.3 pp	29.6×
20	55.2%	100.0%	96.0%	45.7%	−50.3 pp	21.9×
25	55.0%	100.0%	95.0%	45.6%	−49.3 pp	17.3×

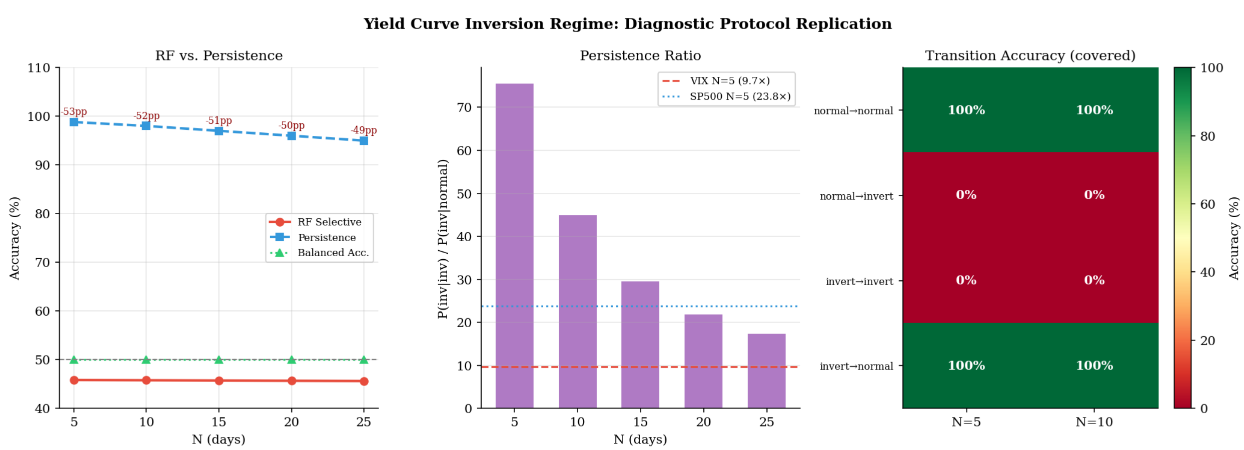

Figure 14: Yield curve inversion regime replication. Left: persistence dominates RF selective at all horizons (RF−Persist = −49 to −53 pp). Center: persistence ratio 75.5× at $N = 5$, the highest across all three tasks. Right: transition accuracy—normal→inverted covered accuracy is 0% and balanced accuracy is 50% at every horizon.

Table 14 and Figure 14 report results. The yield curve inversion result replicates the VIX and S&P 500 findings in the most extreme form yet. The persistence ratio at $N = 5$ is **75.5**× (inverted → inverted: 98.9%; normal → inverted: 1.3%), reflecting the fact that yield curve inversions in the modern rate environment persist for months. Persistence achieves 98.8% accuracy at $N = 5$, while the RF selective classifier achieves only 45.8%—a gap of −53 pp. Notably, 45.8% falls below the majority-class baseline of 54.4%, meaning the RF is performing worse than always predicting the dominant class (inverted); the model is predicting predominantly normal (non-inverted) outcomes despite the inverted days being more common in the test period. Balanced accuracy collapses to exactly 50.0% at every horizon, indicating that once class imbalance is corrected the RF predictions are equivalent to random guessing. Normal→inverted transition covered accuracy is **0.0%** at all horizons, exactly replicating the VIX calm→high and S&P 500 bull→bear findings. The diagnostic protocol identifies the same persistence-illusion mechanism in this macroeconomic regime with even sharper quantitative signatures.

7.4 Interpretation

Table 15 summarizes the key diagnostic statistics across the three replication tasks.

Table 15: Cross-task summary of diagnostic protocol results. Persistence ratio = P(high|high) / P(high|calm) at $N = 5$. All three tasks show 0% transition accuracy, same-regime fraction ≥ 97%, and RF underperforming or barely matching persistence. The persistence ratio predicts the severity of the illusion.

Task	Persist ratio (N=5)	Trans acc	Same-regime %	RF−Persist
VIX regime (main)	9.8×	0%	99%	+0.3 pp (n.s.)
S&P 500 trend-regime	23.8×	0%	99%	−4.0 pp
Yield curve inversion	75.5×	0%	99%	−53.0 pp
Synthetic (oracle features)	4.4×	55%	92%	+8.7 pp

The replication establishes that the persistence illusion is not a VIX-specific artifact. It is a structural consequence of applying selective prediction to any autocorrelated binary time series: the model concentrates its confident predictions on days far from the decision boundary, which are precisely the regime-stable days, and achieves high accuracy there by predicting persistence.

Regime-change days, which are by definition ambiguous and boundary-crossing, fall in the abstention zone or receive confident wrong predictions regardless of the classifier used. Crucially, the severity of the illusion scales with the persistence ratio: a 9.8× ratio (VIX) produces RF−Persist ≈ 0 pp with insignificant bootstrap CI; a 75.5× ratio (yield curve) produces RF−Persist = −53 pp. The synthetic positive control (persistence ratio 4.4×, from Section 6) provides the boundary: when the persistence ratio is moderate and a genuine predictive feature exists, the protocol correctly passes the model. When the persistence ratio is high and no such feature exists in the daily data, the protocol correctly flags the illusion. The diagnostic protocol developed in Section 5 identifies this mechanism consistently across all four settings.

8 Conclusion

This paper applies a six-component diagnostic protocol to VIX regime classification and shows that headline selective prediction accuracy reflects autocorrelation, not machine learning skill. The confidence filter achieves 0% accuracy on every calm→high transition it covers; explicit cost-weighting up to 20× does not rescue this, confirming the failure is informational rather than objective-function-driven. A McNemar test on jointly covered days finds $b = c = 0$ at $N \geq 10$: RF and persistence make identical predictions on every shared day, so the accuracy gap is entirely composition-driven. Three-feature HAR logistic regression matches or exceeds 36-feature RF in point estimate at four of five horizons with no bootstrap-significant gap; cost-weighted utility at $N = 10$ favors persistence whenever false negatives are more costly than false alarms; and all conclusions survive excluding the 2022 bear-market period.

Six supplementary analyses confirm and generalize the persistence-illusion thesis. First, persistence quantification (Section 5.3) shows that 99–100% of the RF's correct covered predictions are same-regime calls at every horizon; at $N \geq 20$, not a single correct prediction is a regime-change call. Second, SHAP attribution (Section 5.7) shows that VIX technical features account for 71.9% of model SHAP mass at $N = 5$—the model overwhelmingly learns VIX moving averages. Third, a two-layer LSTM with 20-day temporal memory (Section 5.5) also fails on calm→high transitions (19–35% accuracy, below random chance), confirming that architectural richness does not resolve the information constraint; event case studies document confident wrong predictions before the August

2024 carry-trade unwind (VIX ≈ 65) and three other crises. Fourth, a synthetic positive control (Section 6) demonstrates that the protocol has power to detect genuine skill: when an event-alarm oracle feature is injected into an otherwise persistence-dominated dataset, the protocol passes the model—RF beats persistence by +8.7 pp at $N = 5$ with calm→high accuracy of 54.9%, satisfying all six diagnostic criteria. Fifth, the diagnostic replicates on S&P 500 trend-regime classification (Section 7): bull→bear covered accuracy is 0%, persistence beats RF at every horizon (gap -4 to -7 pp), and 97–99% of correct RF predictions are same-regime calls—the same pattern at even higher autocorrelation (persistence ratio 23.8× vs. VIX's 9.8×). Sixth, yield curve inversion regime classification (Section 7.3) provides the sharpest case: a persistence ratio of 75.5× produces RF−Persist = −53 pp, confirming that the severity of the illusion scales monotonically with the persistence ratio. Together these replications establish the persistence illusion as a general structural property of selective prediction on autocorrelated labels, not a VIX-specific artifact.

HGB mirrors RF behavior throughout: it achieves similar selective accuracy but also suffers 50% balanced accuracy at $N = 25$ and offers no practical advantage in walk-forward evaluation. The RF's one measurable distinction over persistence is fold-to-fold stability ($N = 5$ fold std: RF 1.7% vs. persistence 3.2%), not directional accuracy. The one exploratory exception is that model abstention carries a small independent increment beyond VIX proximity at $N = 5$ (likelihood-ratio statistic $p = 0.001$ on the evaluation sample, not a confirmatory result), but this is post-hoc and requires dedicated out-of-sample confirmation. The protocol—coverage-matched baseline comparison, regime-transition accuracy audit, persistence quantification, SHAP attribution, sequence-model comparison, and cross-task replication—is modular and applicable to any selective prediction system on serially correlated labels.

Limitations

The feature set consists of publicly available daily market data. Calm→high transitions are disproportionately triggered by exogenous shocks—geopolitical events, policy surprises, sudden liquidity crises—that leave no advance signature in daily price features. In this sense, 0% covered accuracy on transition days is an expected structural property of the problem: the confidence filter fails not by mispredicting knowable patterns but by issuing confident wrong predictions on event-driven days it cannot distinguish from ordinary calm days. The four-year test period (June

2022–May 2026, ≈1,000 trading days) is underpowered for distinguishing accuracy differences of 1–3 pp under serial dependence; a longer out-of-sample window is needed to confirm directional findings. All primary results use a single 80/20 chronological holdout; the walk-forward robustness check (Appendix E) partially addresses this but covers the same calendar period. Cross-task replications each use their own single holdout, so the generalizability claims rest on three independent test windows rather than a systematic multi-window evaluation. The yield curve replication additionally uses a feature set not designed for monetary-policy dynamics; an optimally specified model would include the current T10Y2Y level directly.

Future Work

The primary open direction is feature enrichment: incorporating event-anticipation content such as options skew dynamics, credit spreads, or order-flow imbalance that may carry advance signal about exogenous shocks. The 0% bull→bear accuracy in the S&P 500 replication confirms that this gap is present across asset classes and not addressable by model architecture alone. A two-stage system using the abstention rate as a leading indicator followed by a conditional spike-prediction model is a second natural direction. Third, the diagnostic protocol should be applied to additional autocorrelated regime classification tasks—credit spread regimes, recession prediction, currency crisis indicators—to characterize how the persistence ratio relates to the severity of the selective-prediction bias. The cross-task evidence here (VIX at 9.8×, S&P 500 at 23.8×, yield curve at 75.5×) suggests the relationship is monotone, but the functional form (linear, logarithmic, threshold-based) requires a broader survey. The evaluation infrastructure developed here is directly reusable for any such extension.

A Complete Feature List

Table 16: All 36 engineered features used as model inputs, grouped by category. All features are computed from information available at or before prediction date t.

#	Feature name	Definition	Category
1	VIX_SMA10	10-day simple moving average of VIX	VIX trend
2	VIX_SMA20	20-day simple moving average of VIX	VIX trend
3	VIX_MOM10	$\mathrm{VIX}_t - \mathrm{VIX}_{t-10}$	VIX momentum
4	VIX_MOM20	$\mathrm{VIX}_t - \mathrm{VIX}_{t-20}$	VIX momentum
5	VIX_ROC10	$(\mathrm{VIX}_t/\mathrm{VIX}_{t-10}) - 1$	VIX momentum
6	VIX_ROC20	$(\mathrm{VIX}_t/\mathrm{VIX}_{t-20}) - 1$	VIX momentum
7	VIX_BB_STD	20-day rolling std. of VIX (Bollinger bandwidth)	VIX Bollinger
8	VIX_BB_UPPER	VIX_SMA20 + 2×VIX_BB_STD	VIX Bollinger
9	VIX_BB_LOWER	VIX_SMA20 − 2×VIX_BB_STD	VIX Bollinger
10	VIX_BB_POSITION	(VIX_t − VIX_BB_LOWER) / (4×VIX_BB_STD)	VIX Bollinger
11	VIX_RSI14	14-day Relative Strength Index of VIX	VIX oscillator
12	VIX_MACROSS	VIX_SMA10 − VIX_SMA20	VIX trend
13	VIX_TERM_SPREAD	$\mathrm{VIX3M}_t - \mathrm{VIX}_t$	VIX structure
14	SKEW_LEVEL	Raw CBOE SKEW index value	Tail risk
15	SKEW_CHG5	$\mathrm{SKEW}_t - \mathrm{SKEW}_{t-5}$	Tail risk
16	SKEW_CHG20	$\mathrm{SKEW}_t - \mathrm{SKEW}_{t-20}$	Tail risk
17	SP500_RET1	Log return of S&P 500 over 1 day	Equity
18	SP500_RET5	Log return of S&P 500 over 5 days	Equity
19	SP500_RET20	Log return of S&P 500 over 20 days	Equity
20	SP500_MOM10	S&P 500 price at t − price at $t - 10$	Equity
21	SP500_MOM20	S&P 500 price at t − price at $t - 20$	Equity
22	SP500_RVOL	20-day realized volatility of S&P 500 daily returns	Equity
23	SP500_DRAWDOWN	252-day maximum drawdown of S&P 500	Equity
24	GOLD_RET1	Log return of gold futures over 1 day	Cross-asset
25	GOLD_RET5	Log return of gold futures over 5 days	Cross-asset
26	GOLD_RET20	Log return of gold futures over 20 days	Cross-asset
27	TNY_CHG1	10Y Treasury yield change over 1 day	Cross-asset
28	TNY_CHG5	10Y Treasury yield change over 5 days	Cross-asset
29	TNY_CHG20	10Y Treasury yield change over 20 days	Cross-asset
30	DXY_RET1	Log return of DXY over 1 day	Cross-asset
31	DXY_RET5	Log return of DXY over 5 days	Cross-asset

B Threshold Sensitivity

Table 17 and Figure 15 show RF selective accuracy and coverage across confidence thresholds for $N = 5$ and $N = 10$.

Table 17: Random Forest selective accuracy and coverage as a function of confidence threshold τ (fixed-spec calibrated pipeline).

τ	RF N=5 Acc	N=5 Cov	RF N=10 Acc	N=10 Cov
0.10	86.5%	91.0%	83.3%	90.5%
0.15	88.3%	85.1%	84.7%	85.9%
0.20	89.0%	82.9%	86.6%	78.5%
0.25	91.9%	76.3%	90.0%	60.1%
0.30	92.3%	73.6%	91.8%	53.6%
0.35	95.0%	67.1%	95.1%	36.5%
0.40	97.4%	49.0%	97.4%	15.3%

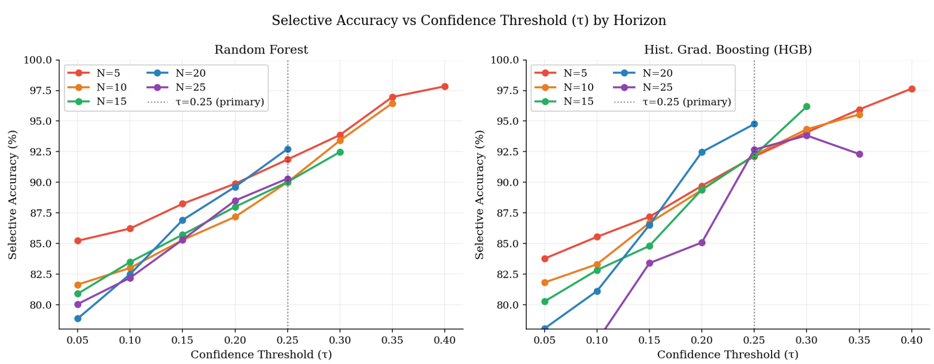

Figure 15: Random Forest selective accuracy as a function of confidence threshold τ for each prediction horizon N. Accuracy increases monotonically with τ across all horizons.

Accuracy increases monotonically with τ across all horizons for the RF, confirming that the calibrated probability is a reliable ranking signal. However, increasing τ does not improve coverage of transition days—it makes confident wrong predictions on calm→high days even more selective (Section 5.4). No choice of τ converts the framework into a regime-change detector.

C Robustness Across VIX Thresholds

Table 18: Random Forest accuracy gain (selective vs. baseline) by VIX regime threshold at $\tau = 0.25$. All gains are positive except RF at $N = 25$ with VIX ≥ 18 (-14.8%), where the gain collapses due to the degenerate prediction behavior at $N = 25$.

N	VIX ≥ 18	VIX ≥ 20	VIX ≥ 22
5	+5.4 pp	+8.5 pp	+7.5 pp
10	+11.3 pp	+10.5 pp	+10.3 pp
15	+14.8 pp	+13.8 pp	+7.9 pp
20	+22.3 pp	+15.9 pp	+9.1 pp
25	−14.8 pp	+16.9 pp	+8.3 pp

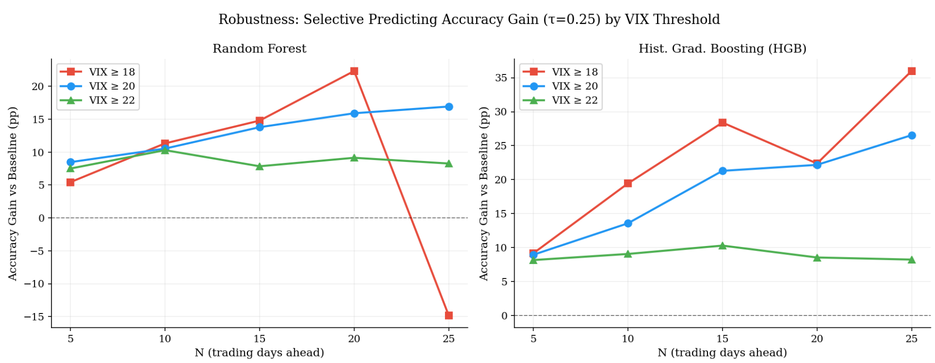

Figure 16: Accuracy gain from selective prediction across VIX thresholds of 18, 20, and 22, plotted against prediction horizon for Random Forest (left) and HGB (right).

The advantage of selective prediction holds across VIX thresholds of 18, 20, and 22 at $N \leq 20$. The exception is RF at $N = 25$, VIX ≥ 18, consistent with the degenerate behavior at the longest horizon (Table 3).

D Sustained Regime Labels

Table 19: Random Forest selective accuracy for instantaneous and sustained regime labels at $\tau = 0.25$.

N	Instantaneous	Sustained
5	91.9%	94.0%
10	90.0%	93.8%
15	90.0%	93.8%
20	92.7%	94.3%
25	90.3%	89.9%

Sustained regime labels (VIX ≥ 20 on every day $t+1$ through $t+N$) are consistently easier to predict than instantaneous labels at $N \leq 20$, as persistent high-volatility episodes carry stronger multi-feature signals. At $N = 25$, the slight reversal reflects the scarcity of sustained positive examples rather than a change in model behavior.

E Walk-Forward Validation

Table 20 reports five-fold expanding walk-forward results for $N \in \{5, 10, 20\}$.

Table 20: 5-fold expanding walk-forward validation. "Mean Baseline" = full-coverage (uncalibrated) accuracy; "Mean Selective" = accuracy at $\tau = 0.25$ with coverage matching. For Persistence, "Baseline" is unconditional persistence accuracy (full coverage) and "Selective" is coverage-matched persistence accuracy at the same threshold as RF. Bold entries indicate where persistence exceeds RF selective accuracy. At $N = 20$, RF/HGB fold standard deviations exceed 25 pp; persistence remains stable (std 7.7%).

Model	N	Mean Baseline	Std	Mean Selective	Std
Random Forest	5	82.8%	9.7%	92.2%	1.7%
Persistence	5	88.0%	3.2%	**93.9%**	3.2%
HGB	5	84.0%	8.0%	84.6%	9.7%
Random Forest	10	72.3%	22.7%	85.9%	7.7%
Persistence	10	84.8%	4.5%	**91.3%**	4.1%
HGB	10	70.9%	22.9%	77.3%	15.2%
Random Forest	20	67.5%	22.8%	67.1%	28.3%
Persistence	20	81.5%	3.9%	**86.2%**	7.7%
HGB	20	63.6%	25.2%	61.4%	29.9%

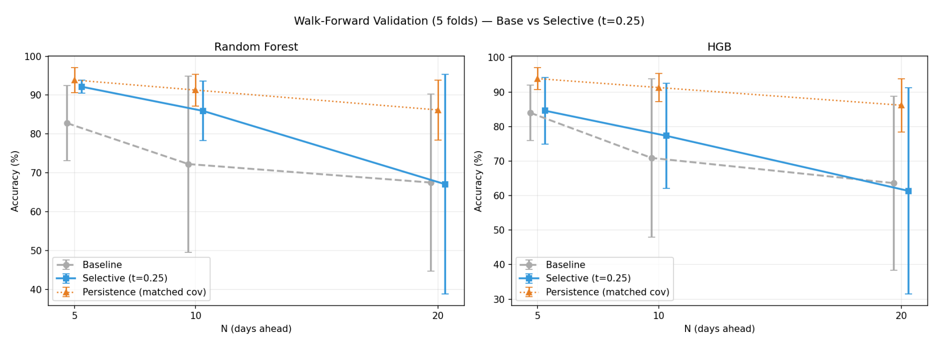

Figure 17: Walk-forward validation results with error bars (±1 std) for baseline, selective prediction, and coverage-matched persistence (orange) across $N \in \{5, 10, 20\}$ for RF (left) and HGB (right). Persistence matches or exceeds RF selective at all horizons.

At $N = 5$, RF selective achieves 92.2% ± 1.7%; the RF's distinctive property is fold stability, not accuracy over persistence (93.9% ± 3.2%). At $N = 10$, persistence (91.3% ± 4.1%) exceeds RF (85.9% ± 7.7%) in both mean and stability. At $N = 20$, RF and HGB collapse (fold std > 25 pp); persistence remains stable. The practical reliable prediction window is $N \leq 10$.

F Probability Calibration Quality

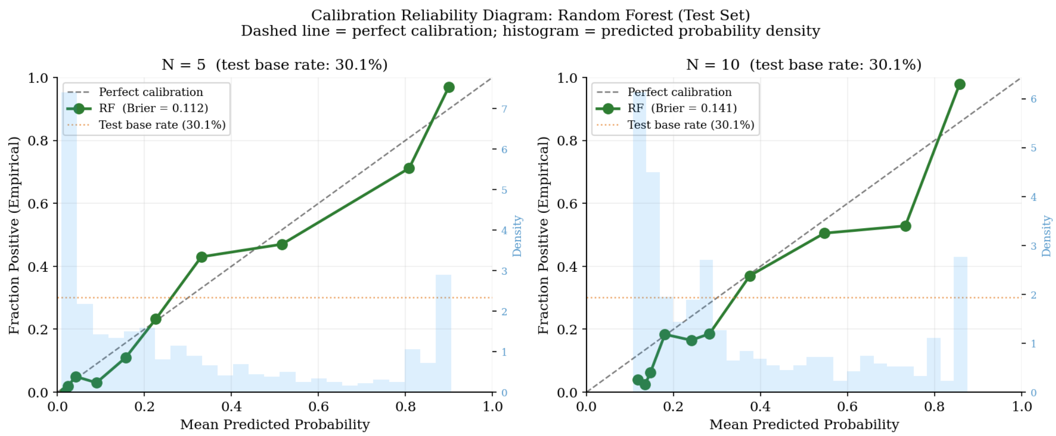

Figure 18: Calibration reliability diagrams for RF at $N = 5$ (left) and $N = 10$ (right). Points near the diagonal indicate well-calibrated probabilities. Orange dotted line = test-set base rate (≈30%).

Calibrated probabilities are broadly well-ordered on the test set. Brier scores are 0.112 at $N = 5$ and 0.142 at $N = 10$, both well below the no-skill baseline $\bar{p}(1 - \bar{p}) \approx 0.21$ at the test-set base rate of 29.8%. Consistent with the base-rate shift (36.7% training, 29.8% test), the calibration curve at mid-range probabilities sits slightly above the diagonal, but the monotonic τ-accuracy relationship in Table 17 is preserved.

G Permutation Feature Importance

Table 21: Top-8 Random Forest features by permutation importance (mean accuracy decrease when feature is shuffled), averaged across $N \in \{5, 10, 20\}$. The remaining 28 features each cause less than 0.15% average accuracy drop.

Rank	Feature	Avg. Importance	Category
1	VIX_SMA10	1.88%	VIX trend
2	SP500_DRAWDOWN	1.08%	Equity risk
3	VIX_BB_LOWER	0.52%	VIX Bollinger bands
4	SP500_MOM10	0.46%	Equity momentum
5	SP500_RVOL	0.31%	Realized volatility
6	SP500_RET20	0.27%	Equity return
7	VIX_BB_POSITION	0.19%	VIX Bollinger bands
8	VIX_ROC20	0.18%	VIX momentum

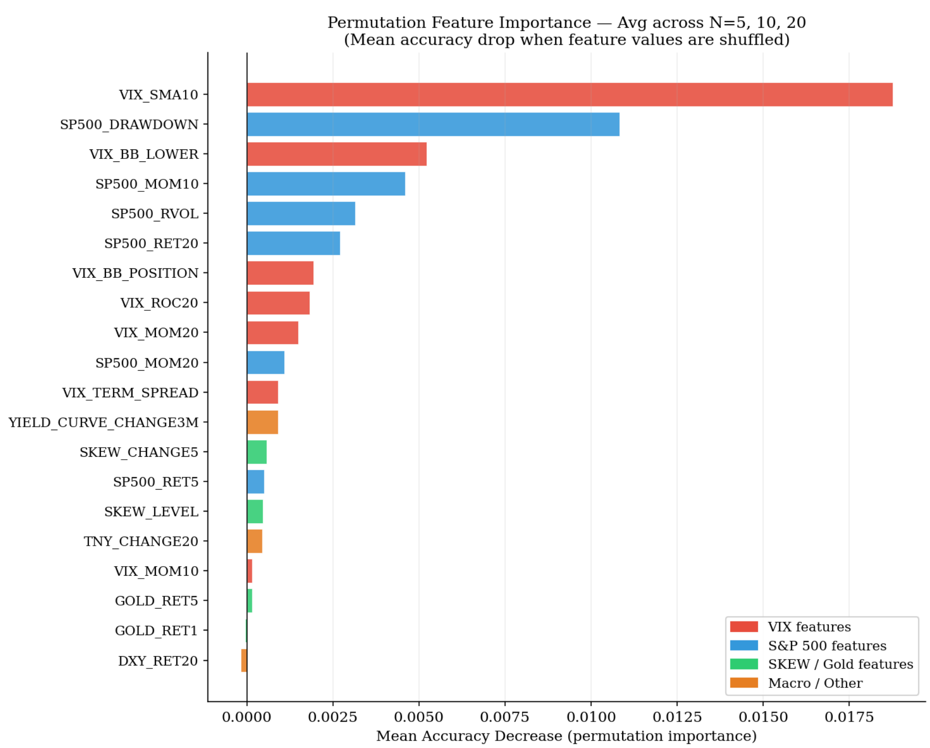

Figure 19: Permutation feature importance averaged across $N = 5, 10, 20$. Avoids the high-cardinality inflation of MDI.

VIX_SMA10 ranks first (1.88% accuracy drop when shuffled); S&P 500 drawdown (SP500_DRAWDOWN) is the highest-ranked non-VIX feature at 1.08%. All values are small in absolute terms (each feature $< 2\%$ accuracy drop), consistent with the SHAP finding that VIX autocorrelation captures most predictable variation. The ranking is qualitatively consistent with the SHAP attribution in Section 5.7.

References

Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794.

Bartlett, P. L. and Wegkamp, M. H. (2008). Classification with a reject option using a hinge loss. *Journal of Machine Learning Research*, 9:1823–1840.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327.

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.

Chow, C. K. (1970). On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16(1):41–46.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196.

Fernandes, M., Medeiros, M. C., and Scharth, M. (2014). Modeling and predicting the CBOE market volatility index. *Journal of Banking & Finance*, 40:1–10.

Geifman, Y. and El-Yaniv, R. (2017). Selective classification for deep neural networks. *Advances in Neural Information Processing Systems*, 30.

Lundberg, S. M. and Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.

Kunsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3):1217–1241.

Guidolin, M. and Timmermann, A. (2007). Asset allocation under multivariate regime switching. *Journal of Economic Dynamics and Control*, 31(11):3503–3544.

Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384.

Whaley, R. E. (2009). Understanding the VIX. *Journal of Portfolio Management*, 35(3):98–105.